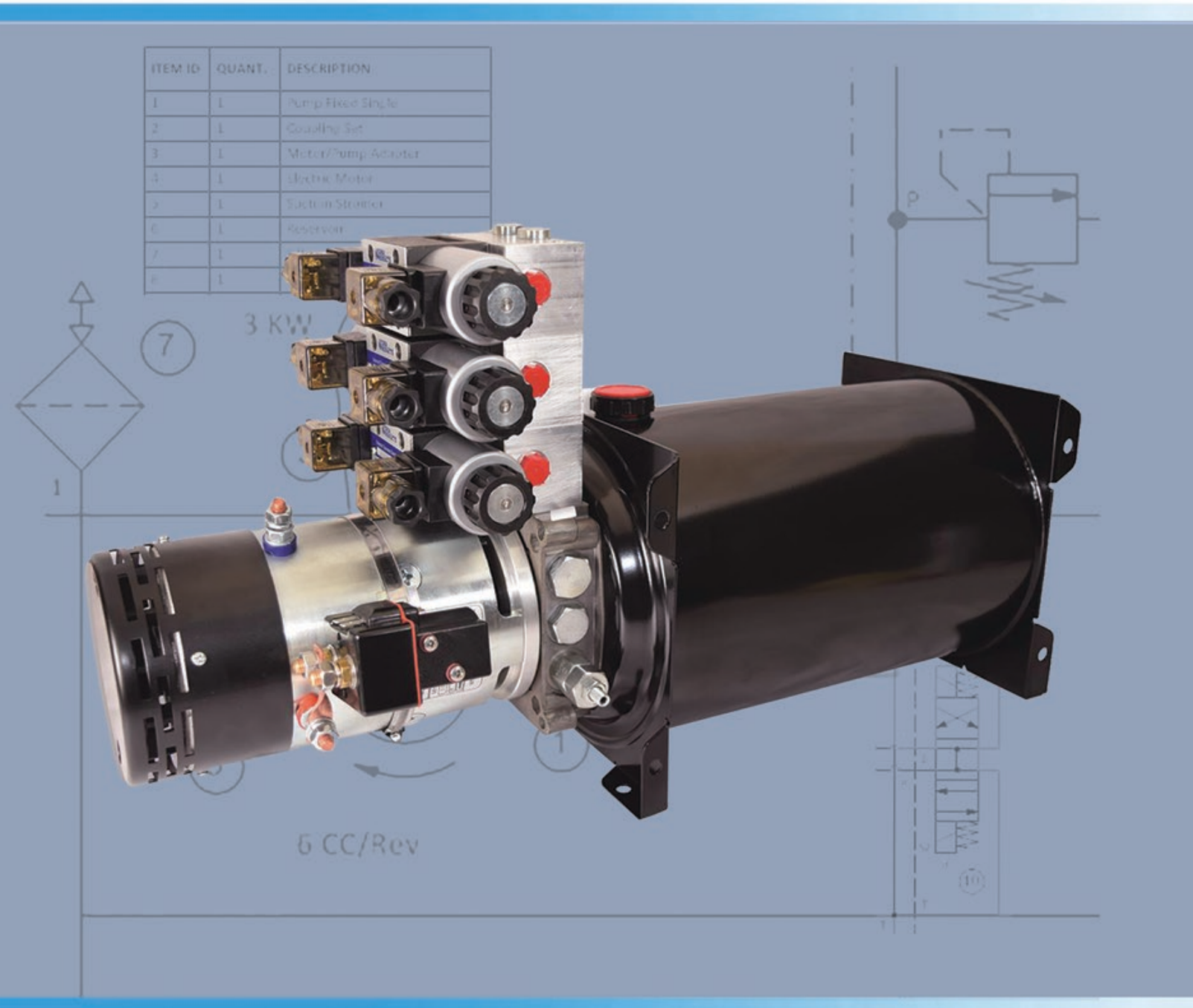
MADE IN BRITAIN



HPM Mini Powerpacks

www.hydraproducts.co.uk

HYDRA PRODUCTS






Micro Hydraulic Power Pack

The Hydraproducts Micro Hydraulic Power Pack is a compact unit, ideal for environments where space is a major issue.



- *Single or Double acting*
- *240V and 415V AC motors*
- *12V and 24VDC motors to 800Watt*
- *Pressures to 200bar*
- *High pressure gear pumps to 1.5cc/rev*
- *Reservoirs to 1.5 Litres*

Mini Hydraulic Power Pack

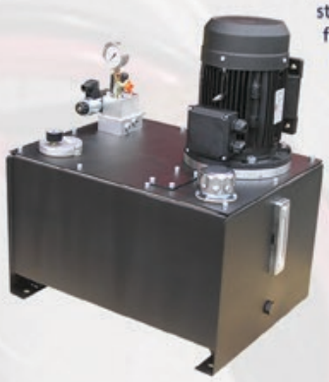


Hydraproducts AC & DC powered Mini Power Packs are suitable for any Mobile, Industrial or Marine application.

- *AC and DC specifications available*
- *DC motors from 500W to 3kW, in 12V and 24VDC*
- *Pressures to 200bar*
- *AC motors from 0.25 to 5.5kW*

Standard Hydraulic Power Unit

The Standard Hydraulic Power Unit uses a standard range of modular components for easy specification, assembly and maintenance in the most demanding industrial applications.



- *Motors up to 30kW*
- *Flow rates to 100 Litres/minute*
- *Pressure capability to 250bar*
- *Tank capacities to 180Litres*

NEW Components Division



- *Valves*
- *Bellhousings & Couplings*
- *Accumulators*
- *Manifold blocks*
- *Coolers*
- *Tanks*
- *Cylinders*
- *Filters*
- *Pumps*

MEMBER OF:

BFPA

REPRESENTING HYDRAULIC & PNEUMATIC INDUSTRIES



HPM Mini Pack

HPM Mini Packs modular design offers a flexible basis for a wide range of mobile and industrial hydraulic systems. Available for AC or DC power from 0.18 to 5.5 kW in single and double-acting formats with flows up to 30 litres/minute and pressures of 320 bar.

Typical Mini Pack applications include:

- Waste Compactor/Balers
- Tipping Trucks and Trailers
- Lifts (Direct and Scissor)
- Road Blockers and Bollards
- Industrial machines

Hydraproducts have a proven record of performance.

- 100% testing
- Wide application knowledge
- Large stock holding
- Fast deliveries
- Product support
- Ability to tailor designs
- Global customer base
- 3D modelling facilities

HPM Mini Pack

HPM Mini Aggregate bieten einen modularen Aufbau, eine flexible Basis und eine breite Palette von mobilen und industriellen Hydrauliksystemen. Verfügbar für AC-oder DC-von 0,18 bis 5,5kW in Einzel-und doppelt wirkende mit Systems Strömungen bis 30l/min und Drücken von 320 bar.

Typische Anwendungen sind

- Müllpresse/Pressen
- Kippenhanger
- Scherenhubhische
- Sicherheits Poller
- industrielle Maschinen

Hydraproducts haben eine nachgewiesene reputation.

- Erfahrung in vielen Anwendungen bereiche
- Kurze Lieferzeiten
- Großes Teilelager
- Kundenservice
- Kundenspezifische Produkte
- 3D Produkt design

HPM Mini Pack

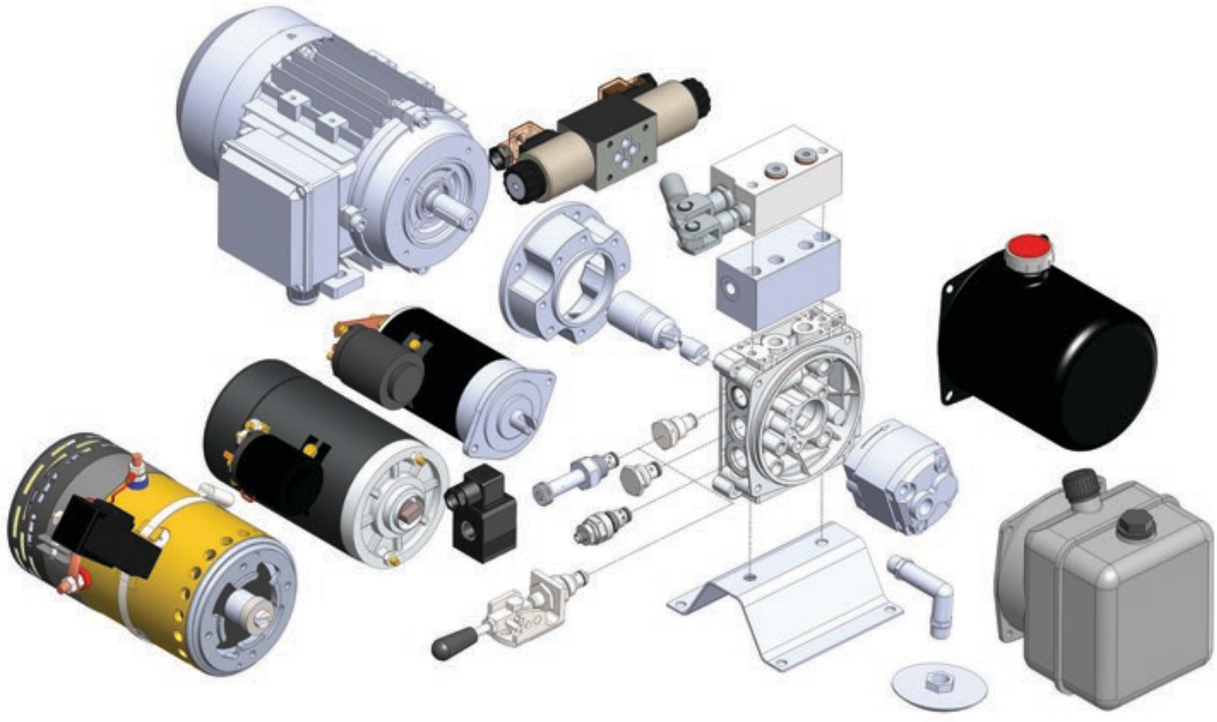
HPM Mini Packs diseño modular ofrece una base flexible para una amplia gama de sistemas hidráulicos industriales y móviles. Disponibles para la energía AC o DC desde 0,18 hasta 5,5 kW en formato simple y doble, con caudales de hasta 30 litros / minuto y presiones de 320 bar.

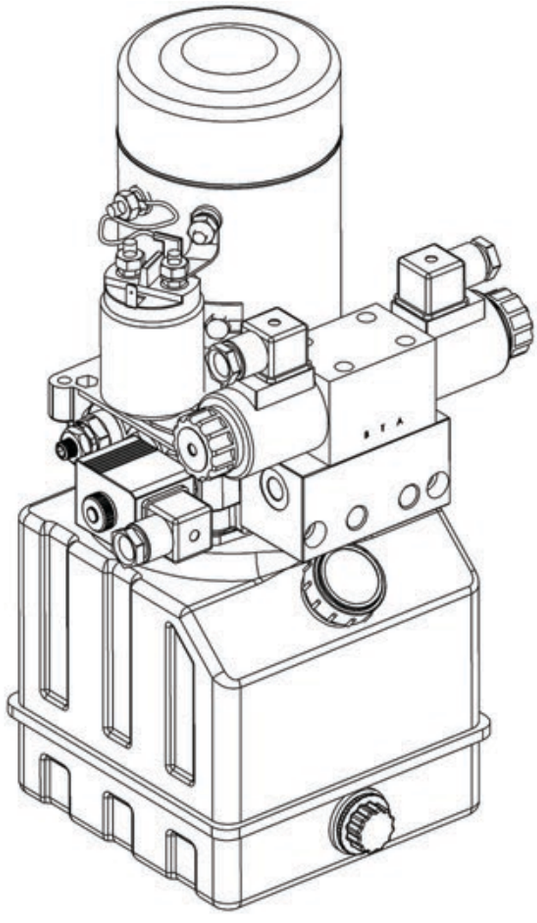
Las aplicaciones típicas incluyen paquete de Mini:

Compactador de residuos / Empacadoras
Camiones y remolques de vuelco
Ascensores (directa y de tijera)
Bloqueo de carreteras y bolardos
máquinas industriales

Hydraproducts tienen un historial comprobado de desempeño.

100% de las pruebas
Conocimiento de las aplicaciones de ancho
Amplio stock celebración
entregas rápidas
de soporte de producto
Capacidad de adaptar los diseños
Base global de clientes
Las instalaciones de modelado en 3D





Specifications

Pump

Type	Gear pumps
Size	0,26 to 9,8 cc/rev

Motors

12/24/48/72VDC	0.5 to 3kW
240VAC	0.25 to 3kW
415VAC	0.25 to 5.5kW

Hydraulic circuits

Wide variety of single acting and double acting circuits

Reservoirs

Plastic	1.5 to 8 Litre
Steel	1.5 to 40 Litre
	Bespoke designs on request

Performance

Pressure	to 320bar
Flow	to 30lpm

Hydraulic oil

Recommended—Mineral Oil HLP HV 6743/4 DIN 51519

Viscosity ISO3448/VG32 (20-100 permissible)

HPM Mini power pack range

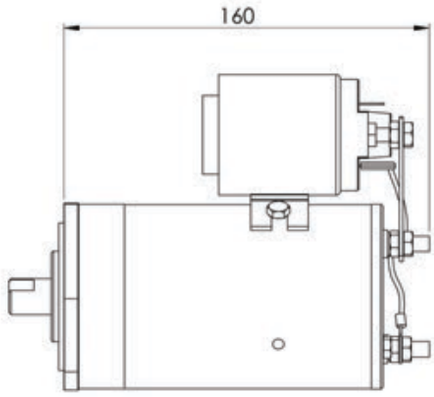
Example of coding → HPM

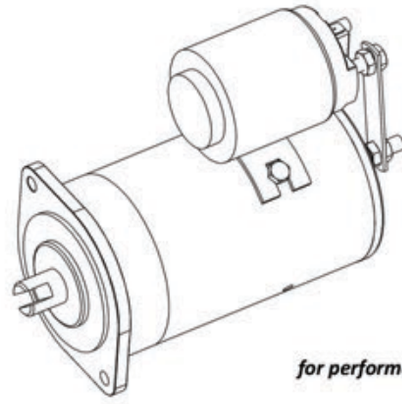
DC Motor Range AC Motor Motor connection flanges and drive couplings	TD4
"W" Gear Pumps "H" Gear Pumps Group 0 Gear pumps and adaptors	33W
Orientation	H
Central Manifold - Types and Circuits	S
Electric Solenoid Cartridge Valve Manual Lever Cartridge Valve Control Valve Voltage	C12D
Flow Control Insert	N
Reservoirs , Steel and Plastic	05P
NG6 (Cetop03) Control Valve Plates	N
NG6 (Cetop03) Directional Control Valves	N
Customer build number	4567

DC MOTORS

TD4

DC Motors— 0.5kW and 0.8kW IP44 - *not supplied with a starter switch (see below)*

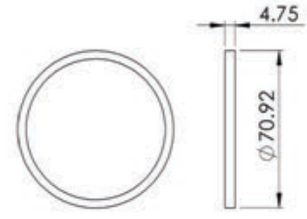




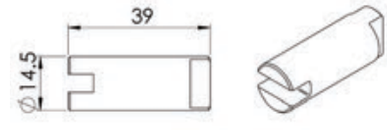
for performance data see page 14

Description	*Ordering code*	*pack code*
12VDC 0.5kW	HP4143	105
12VDC 0.8kW	HP4145	108
24VDC 0.5kW	HP4144	205
24VDC 0.8kW	HP4146	208

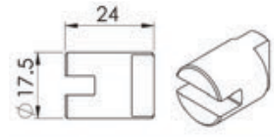
500/800W 80mm Motor Ring	HP050109



500W/800W 80mm Grp 0 Coupling	HP050486

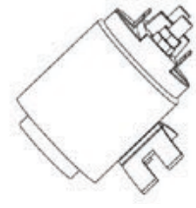


500W/800W 80mm Grp 1 Coupling	HP150027



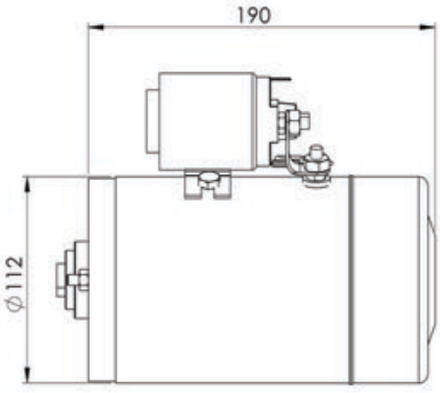
DC Motors— Start Switch

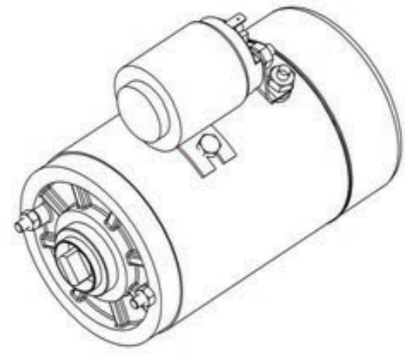
12v 200 Amp start switch	HP4454-12
24v 150 Amp Start switch	HP4454-24



DC MOTORS

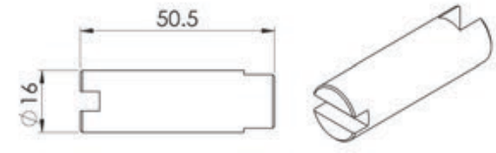
DC Motors— 1.8kW and 2kW with start switch and thermal protection IP54



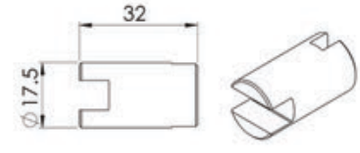


Description	Ordering code	pack code
12VDC 1.8kW	HP4421-12V	120
24VDC 2kW	HP4421-24V	220

1.8-2.2kW 112mm to Grp 0 Coupling	HP050252



1.8-2.2kW 112mm to Grp 1 Coupling	HP050130



DC Motors— Plastic Clip on Cover *(for HP4421 motors only)*

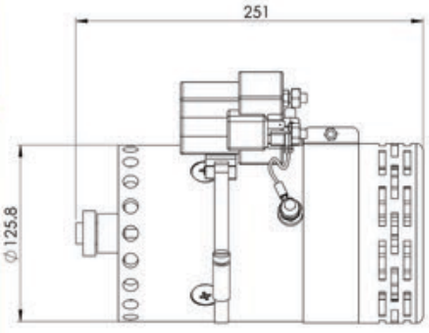
Motor cover	HP150034

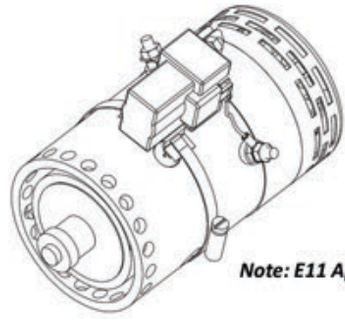


DC MOTORS

TD4

DC Motors — 3kW 24VDC Fan Cooled IP24 - *not supplied with a start switch—see below*



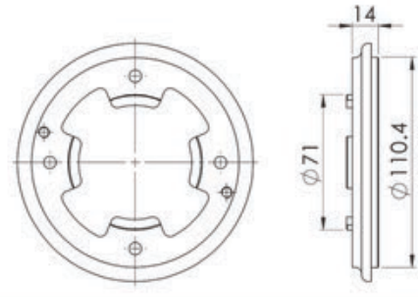


for performance data see page 16

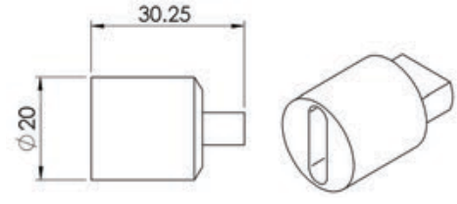
Note: E11 Approved when fitted with SW80 Start Switch

Description	*Ordering code*	*pack code*
24VDC 3kW	EC3/2.4/35.15	41F

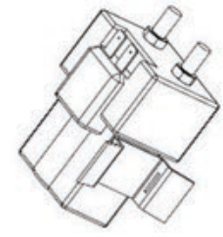
125mm Motor Flange	HP050575



125mm 5” to Grp 1 Coupling	HP4166

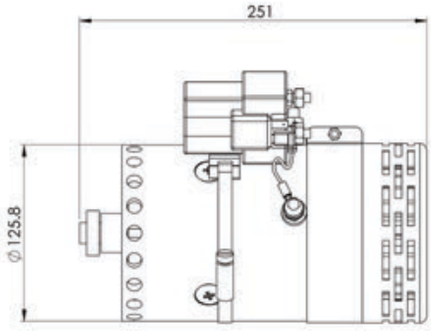


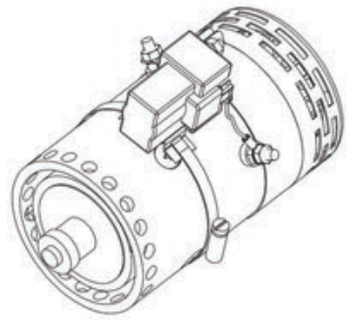
24V 250 Amp Heavy Duty Start Switch	SW80



DC MOTORS

DC Motors — 48VDC Fan Cooled IP24 - *not supplied with a start switch—see below*

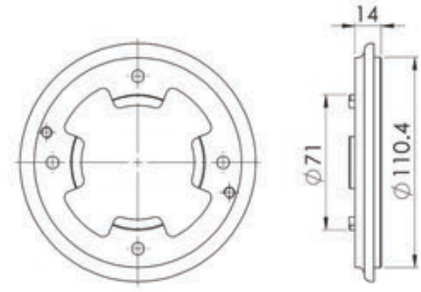




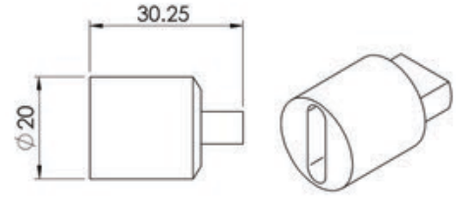
for performance data see page 34

Description	*Ordering code*	*pack code*
48V 1kW	HP050017	41F

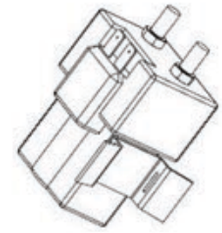
125mm Motor Flange	HP050575



125mm to Grp 1 Coupling	HP4166



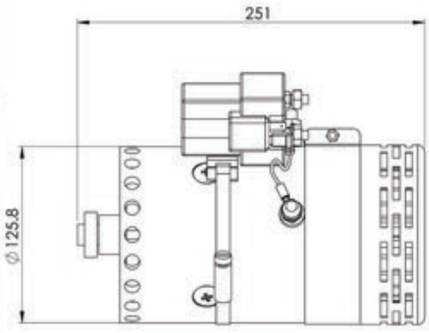
48V Heavy Duty Start Switch	SW80—2075P

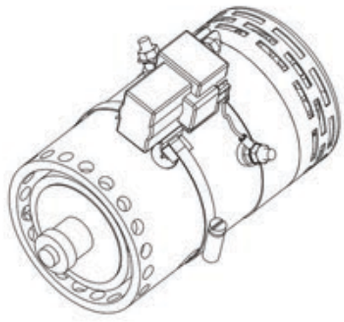


DC MOTORS

TD4

DC Motors — 72VDC Fan Cooled IP24 - *not supplied with a start switch*

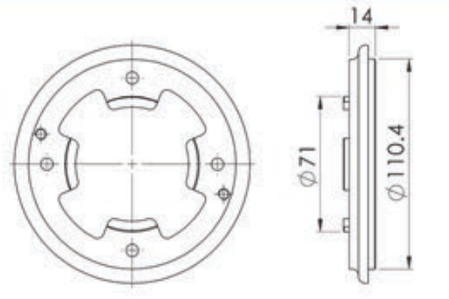




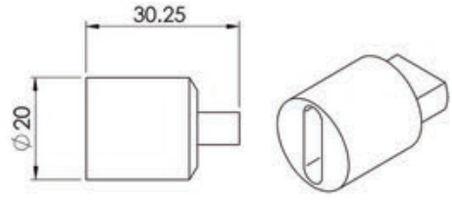
for performance data see page 34

Description	*Ordering code*	*pack code*
72VDC 1kW	HP050018	71F

125mm Motor Flange	HP050575



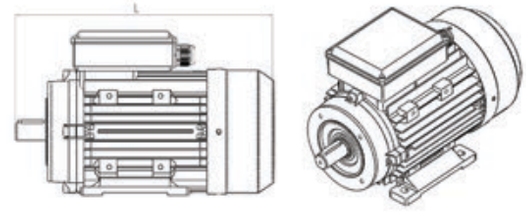
125mm to Grp 1 Coupling	HP4166



AC MOTORS

AC Motors — 1 phase 240V 50Hz B34 IEC

Description	Length	Ordering code	Pack code
0.37kW 2pole 71 frame	180	MYSC2	SC2
0.55kW 2pole 71 frame	180	MYSD2	SD2
0.75kW 2pole 80 frame	250	MYSE2	SE2
1.1kW 2pole 80 frame	250	MYSF2	SF2
1.5kW 2pole 80 frame	250	MYSG2	SG2
2.2kW 2pole 90 frame	285	MYSH2	SH2
3.0kW 2pole 100 frame	315	MYSI2	SI2
0.37kW 4pole 71 frame	180	MYSC4	SC4
0.55kW 4pole 80 frame	180	MYSD4	SD4
0.75kW 4pole 80 frame	250	MYSE4	SE4
1.1kW 4pole 90 frame	285	MYSF4	SF4
1.5kW 4pole 90 frame	285	MYSG4	SG4
2.2kW 4pole 100 frame	285	MYSH4	SH4
3.0kW 4pole 100 frame	285	MYSI4	SI4



MY Permanent capacitor motors are preferred with a 0.5-0.9 FLT starting torque.

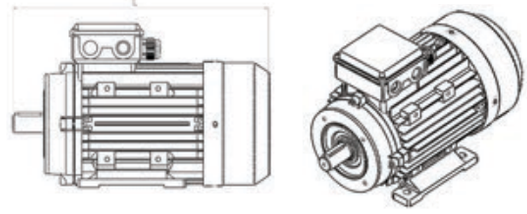
MC Capacitor start/run motors have a high starting torque.

All motors are aluminium body with a multi-mount feet. Insulation Class F, IP55 protection.

IP65 options available

AC Motors — 3 phase 380/415V 50Hz B34 IEC

0.37kW 2pole 71 frame	211	MSTC2	TC2
0.55kW 2pole 71 frame *	225	MSTD2	TD2
0.75kW 2pole 80 frame *	225	MSTE2	TE2
1.1kW 2pole 80 frame	250	MSTF2	TF2
1.5kW 2pole 80 frame *	250	MSTG2	TG2
2.2kW 2pole 90 frame	317	MSTH2	TH2
3kW 2pole 100 frame	326	MSTI2	TI2
4kW 2pole 100 frame *	327	MSTJ2	TJ2
5.5kW 2pole 112 frame	335	MSTK2	TK2
0.37kW 4pole 71 frame	211	MSTC4	TC4
0.55kW 4pole 71 frame *	225	MSTD4	TD4
0.75kW 4pole 80 frame	250	MSTE4	TE4
1.1kW 4pole 80 frame	250	MSTF4	TF4
1.5kW 4pole 90 frame	287	MSTG4	TG4
2.2kW 4pole 90 frame *	317	MSTH4	TH4
3.0kW 4pole 100 frame	325	MSTI4	TI4
4.0kW 4pole 100 frame *	331	MSTJ4	TJ4
5.5kW 4pole 112 frame	335	MSTK4	TK4



Dual European voltages, up to 3kW 230/400 above 400/690V. Dual 50/60Hz rated.

S1 Continuous Duty S1. Thermistors PTC.

All motors are aluminium body with a multi-mount feet. Insulation Class F, IP55 protection.

IP65 and Explosion proof options available

** High Output motors in a "reduced" frame size*

TD4

TD4

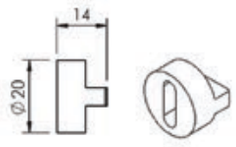
AC MOTOR FLANGES AND COUPLINGS

AC Adaptor Flange and Drive Coupling for 70 Frame B4 IEC Motors

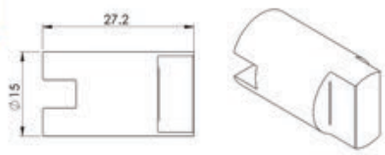
Description	*Ordering code*
71 Frame Adaptor Flange	HP150030-71
71 Frame Coupling	HP050124-71



Drive Dog for Grp1 pumps	HP050131

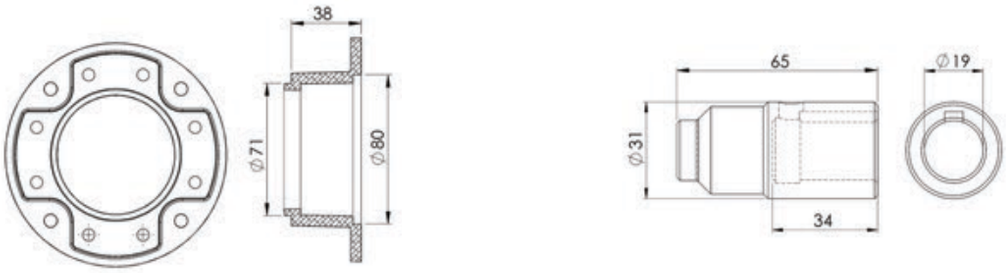


Grp0 Coupling	HP050049

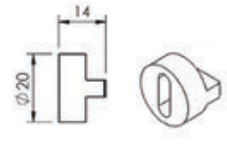


AC Adaptor Flange and Drive Coupling for 80 Frame B4 IEC Motors

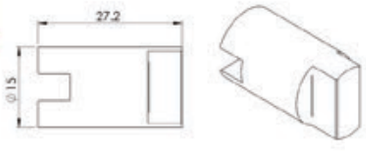
80 Frame Adaptor Flange	HP050312-80
80 Frame Coupling	HP050124-80



Drive Dog for Grp1 pumps	HP050131



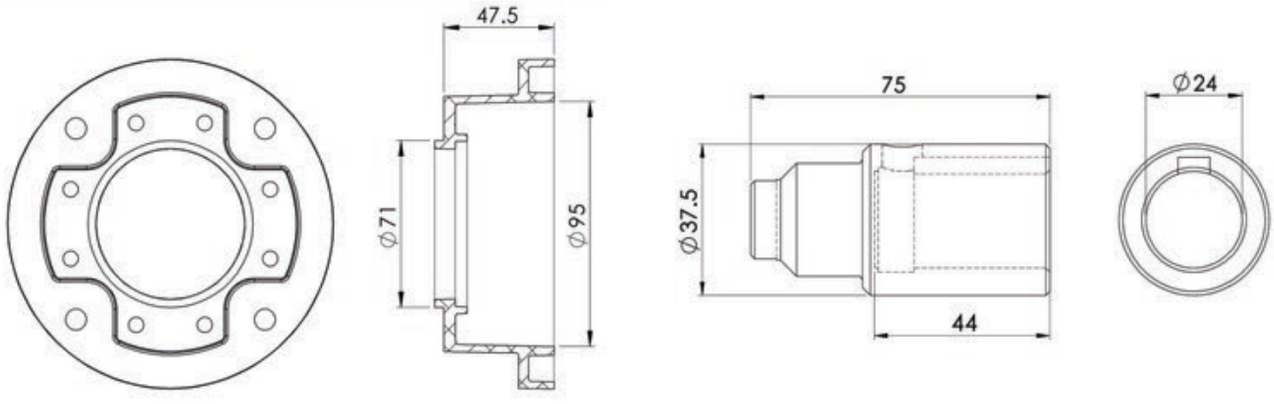
Grp0 Coupling	HP050049



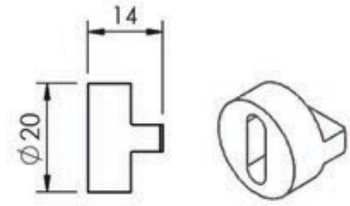
AC MOTOR FLANGES AND COUPLINGS

AC Adaptor Flange and Drive Coupling for 90 Frame B4 IEC Motors

Description	*Ordering code*
90 Frame Adaptor Flange	HP050312-90
90 Frame Coupling	HP050124-90

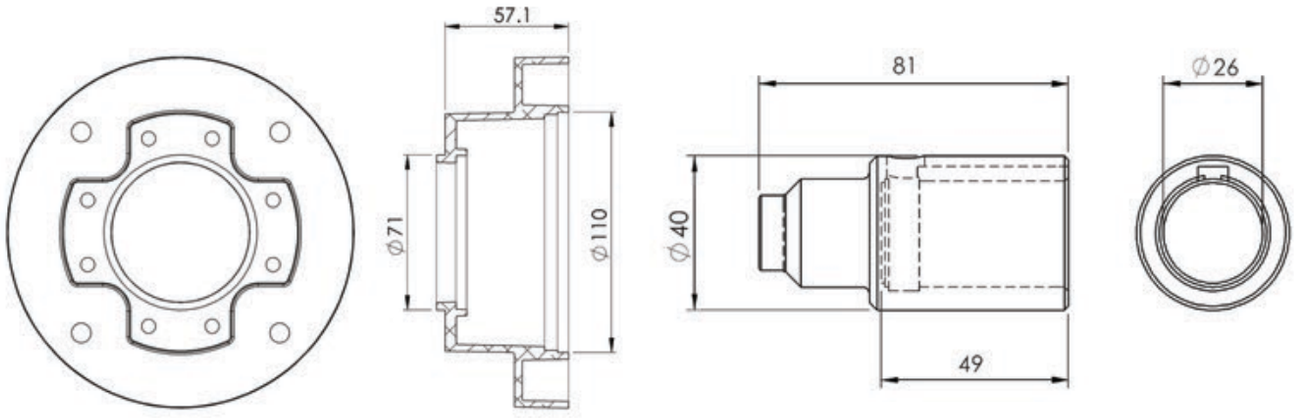


Drive Dog for Grp1 pumps	HP050131

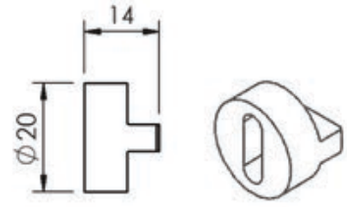


AC Adaptor Flange and Drive Coupling for 110/112 Frame B4 IEC Motors

100/112 Frame Adaptor Flange	HP050312-100
100/112 Frame Coupling	HP050124-100/112



Drive Dog for Grp1 pumps	HP050131

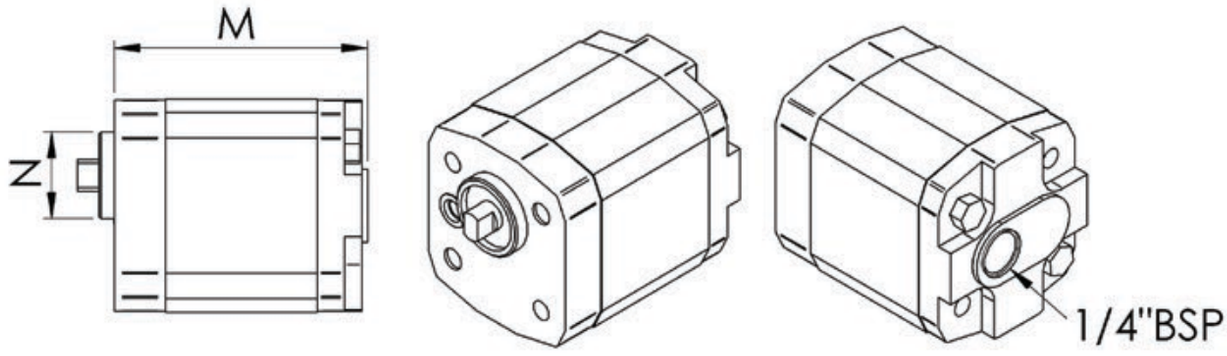


HYDRAULIC PUMPS

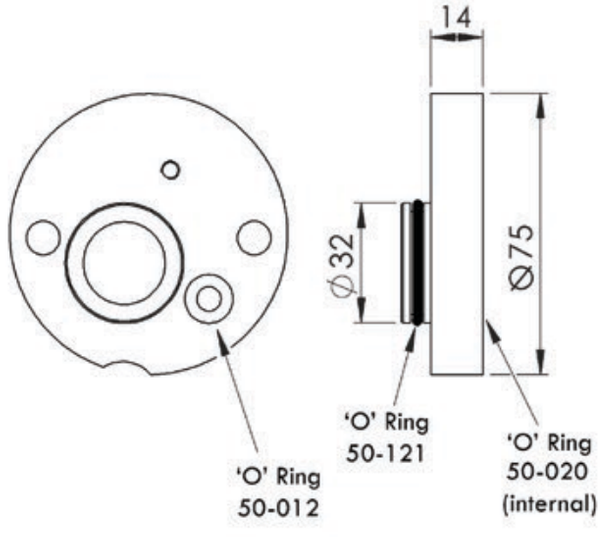
Gear Pump group 0

33W

Size cm^3	P(nom)	P(max)	M(mm)	N(mm)	Ordering code	Pack code
0.25	220	260	60.75	22	HP150046-02C	02C
0.50	220	280	63	22	HP150046-05C	05C

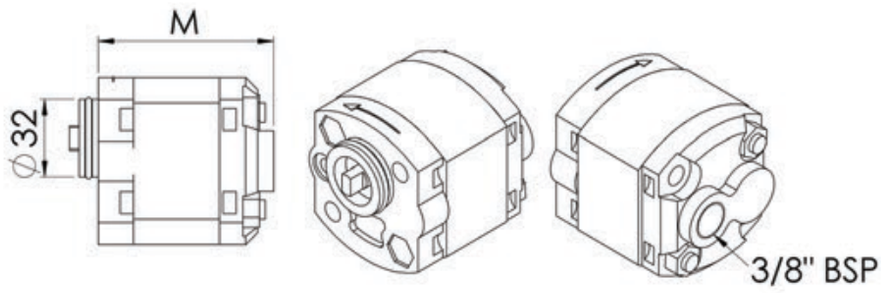


Description	Ordering code
Group 0 Pump Adaptor Flange	HP150028



HYDRAULIC PUMPS

Gear Pump group 1—W Type



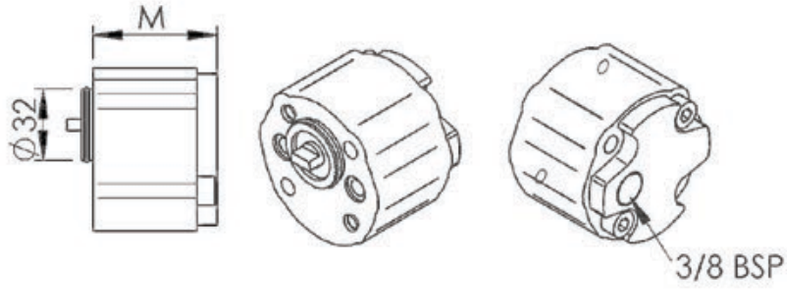
The "W" pump is the standard option for the HPM mini packs with pressures of 210bar.

33W

Size cm^3	P(nom)	P(max)	M(mm)	Ordering code	Pack code
0.7	210	230	73	HP4404_07W	07W
1.0	210	230	74	HP4404_10W	10W
1.3	210	230	75.2	HP4404_13W	13W
1.6	210	230	77	HP4404_16W	16W
2.0	210	230	78.5	HP4404_20W	20W
2.5	210	230	80	HP4404_25W	25W
3.3	210	230	83	HP4404_33W	33W
4.2	210	230	87	HP4404_42W	42W
4.8	200	220	89.5	HP4404_48W	48W
5.8	160	180	94.5	HP4404_58W	58W

HYDRAULIC PUMPS

Gear Pump group 1—H Type for Pressures up to 320 bar

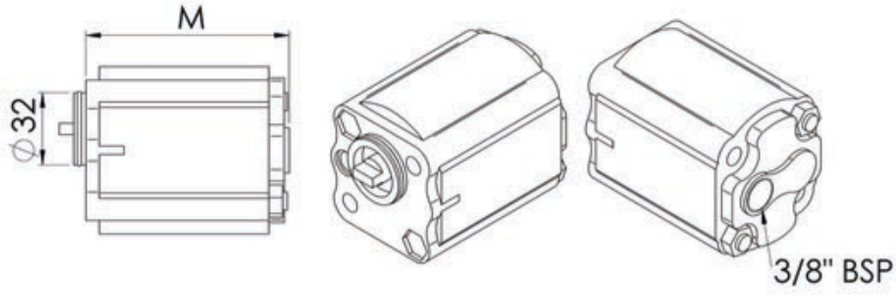


Group 1 "H" type pump gives high pressures to 320 bar (max).

33W

Size cm^3	P(nom)	P(max)	M(mm)	Ordering code	Pack code
1.0	300	320	52.2	HP150035-10	10H
1.2	300	320	53.5	HP150035-12	12H
1.7	300	320	55.3	HP150035-17	17H
2.0	300	320	56	HP150035-20	20H
2.6	300	320	58.7	HP150035-26	26H
3.2	300	320	60.9	HP150035-32	32H
3.8	300	320	63.1	HP150035-38	38H
4.2	300	320	63.4	HP150035-42	42H

Gear Pump group 1—Q Type for "silent" operation



Group 1 "Q" type pump is a Helical gear pump with noise levels typically 10dBA lower than other gear pumps.

Size cm^3	P(nom)	P(max)	M(mm)	Ordering code	Pack code
2.6	250	280	76.0	HP150020-26Q	26Q
4.2	250	280	87.6	HP150020-42Q	42Q
5.7	190	210	93.2	HP150020-57Q	57Q

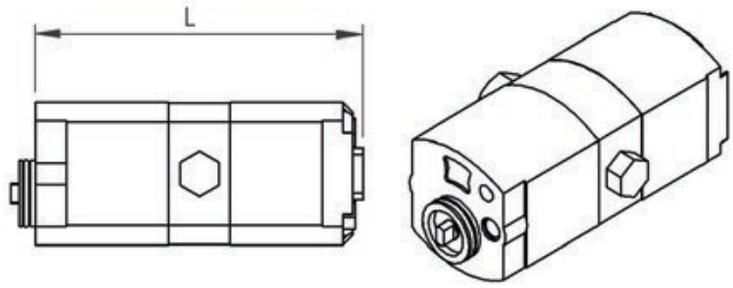
HYDRAULIC PUMPS

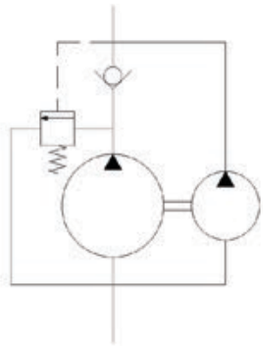
Tandem Gear Pump group 1—Hi-Lo Pump

Description	Length (L)	Ordering code	Pack code
1.2cc+5.0cc with unloading valve	168.3	HP150023-1250T	1250T
3.2cc+7.8cc with unloading valve	185.4	HP150023-3278T	3278T

Tandem pumps have a built-in valve to unload the rear pump at low pressure, the high pressure front pump is set by the power pack relief valve.

33W



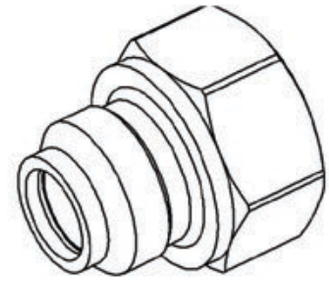


HYDRAULIC PUMP ACCESSORIES

Pump Accessories

Description	Flow	Ordering code
Buffer Valve	5—10 LPM	HCF04-02-00
Buffer Valve	10—15 LPM	HCF04-22-00

By-pass valve enables the pump to start off-load when using a Permanent capacitor motor. 3/8" BSP., these can be fitted to "W " series pumps ONLY.



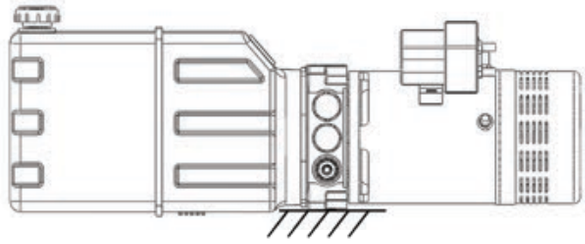
MOUNTING

Orientation

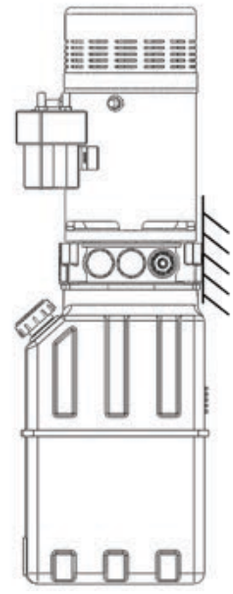
Position	Pack code
Horizontal	H
Vertical	V
Bulkhead	B
Underneath	U

H

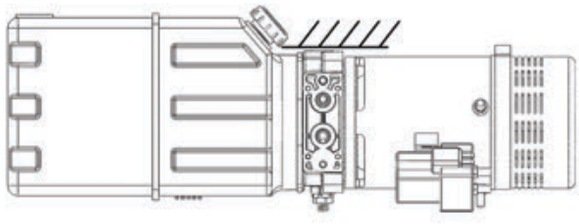
Horizontal



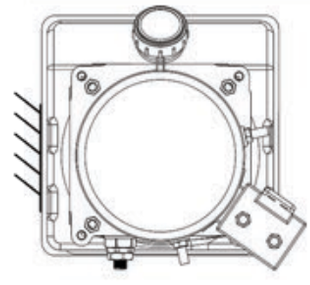
Vertical



Underneath



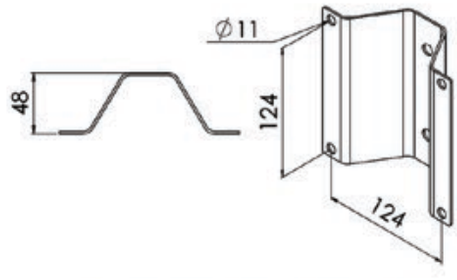
Bulkhead



ACCESORIES

Mounting Bracket

Description	ordering code
Mounting Bracket	HP150015



Stainless Steel Mounting Bracket

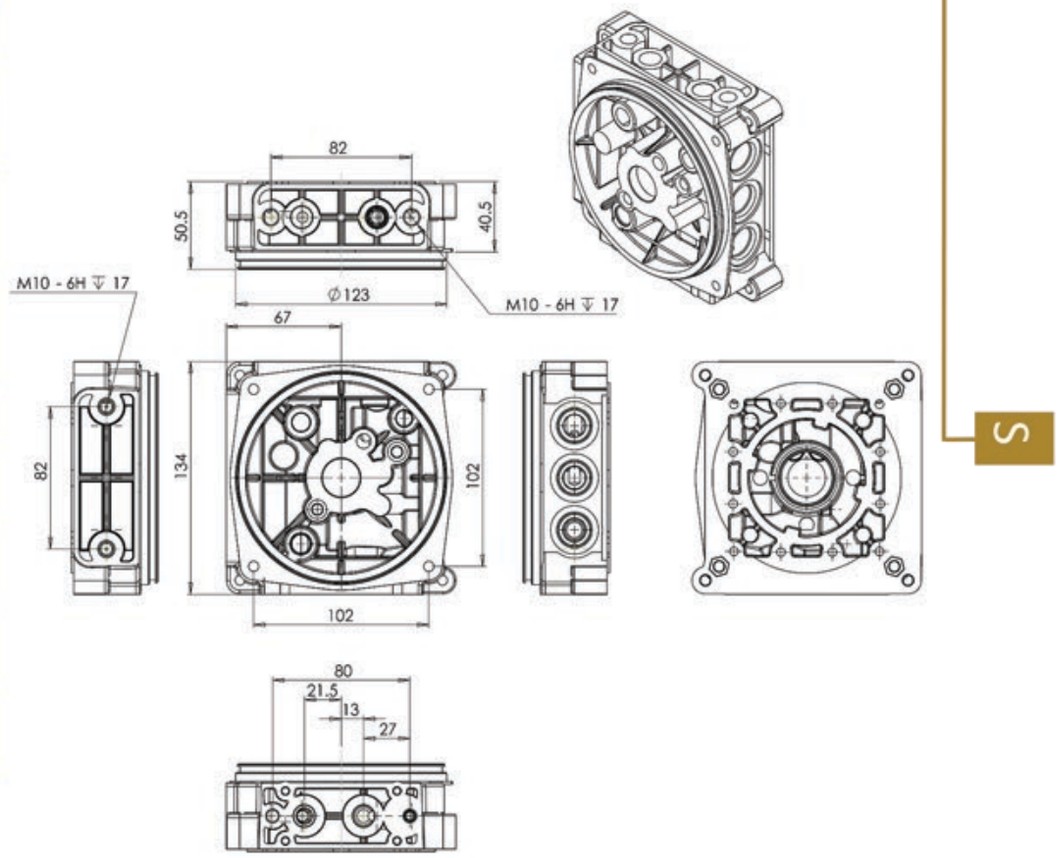
CENTRAL MANIFOLD

HY Central Manifold

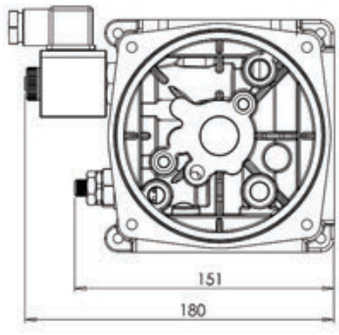
Description	*Ordering code*	*pack code*
*P and T with 1/4"*BSP threads	HP050044	S
P and T with 3/8"BSP threads	HP050045	R

S

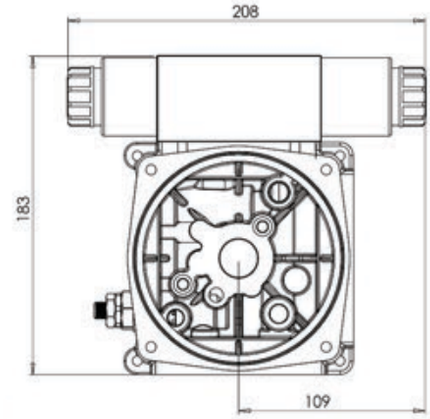
Cavity	Thread
1,2,3	3/4-16 UNF – SAE08
P—T	1/4BSP or 3/8"BSP
Manifold mounting	M8x14
Tank fixing	M6
Pump fixing	M8
Motor fixing-Square	M8
AC Flange fixing	M6
DC Motor fixing	M6



Single Acting Solenoid Valve

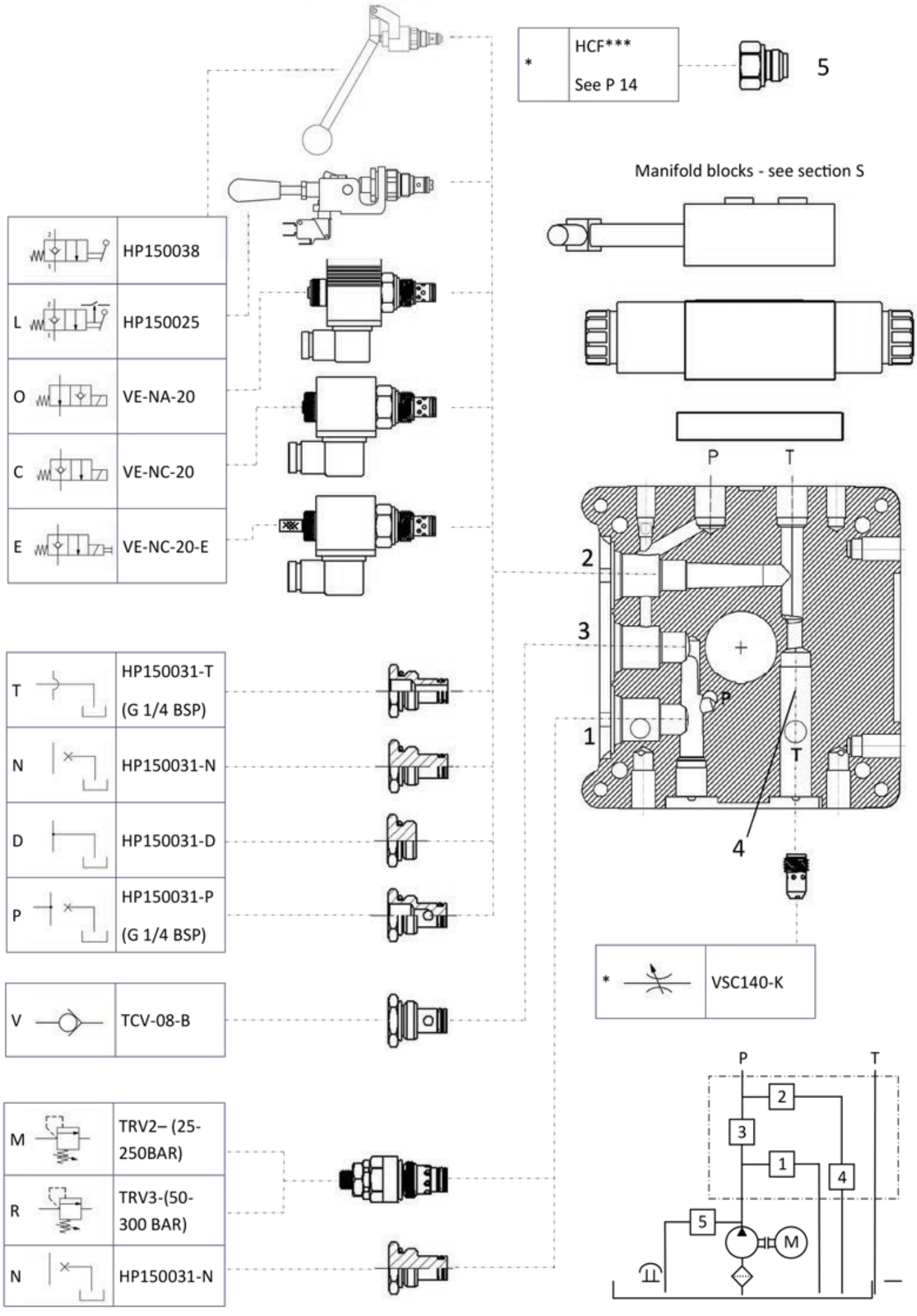


Double Acting with NG6 Valve



Typical Dimensions

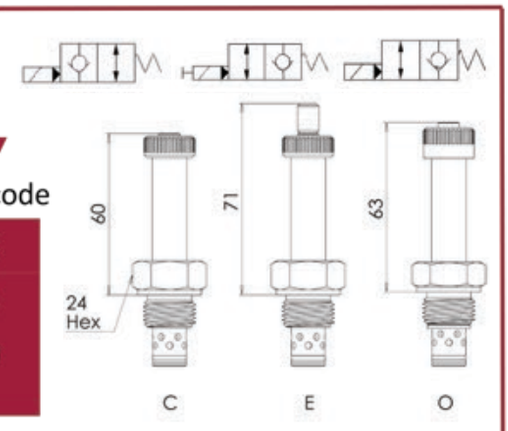
CENTRAL MANIFOLD—HY SIMPLE CIRCUIT



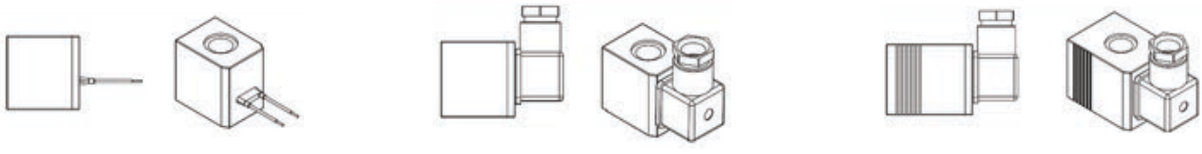
Single Acting Cartridge Valve

Solenoid Operated

Description	order code	pack code
Normally Closed Poppet	VE-NC-20-011N-3/4UNF	C
N/C Poppet + E/Override	VE-NC-20-011N-E-3/4UNF	E
Normally Open Poppet (use 22Watt coils only)	VE-NA-20-O11A-3/4UNF	O

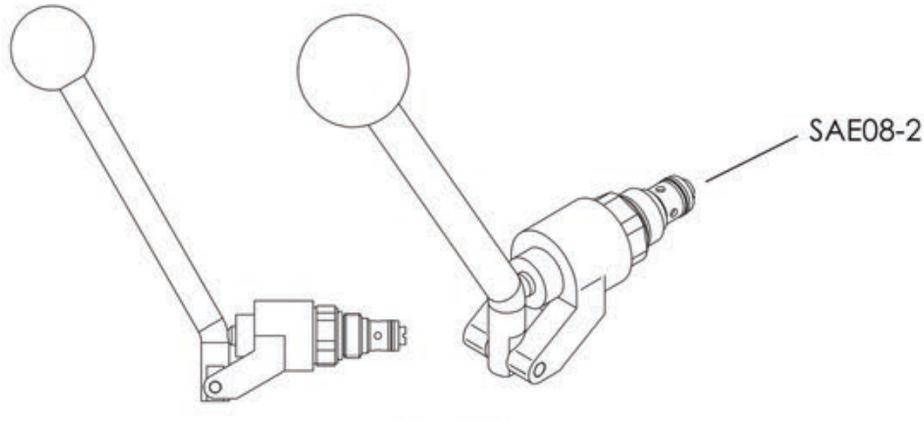


Description	order code	Voltages
18Watt Coil with flying leads	A07-112/124VDC 18W	12VDC, 24VDC
18Watt Coil with DIN connector	EC30-130-12VDC 18W	12VDC, 24VDC, 24VAC, 110VAC,
22Watt Coil with DIN connector	EC36-130-12VDC 22W	12VDC, 24VDC, 24VAC, 110VAC, 240VAC



Options

Description	order code
Manually operated unloading valve	HP150038

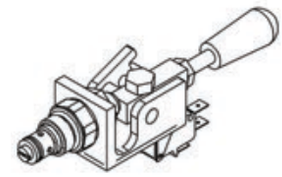


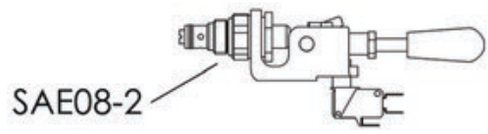
HP150038
LOWERING VALVE

Single Acting Cartridge Valve

Lever Operated

Description	order code	pack code
Normally Closed with Microswitch	HP150025	L

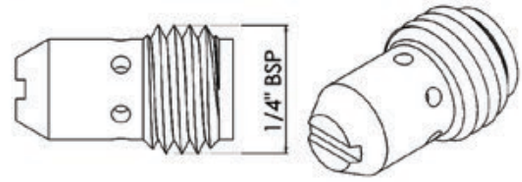


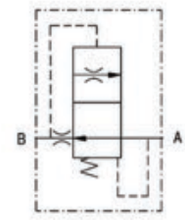


Flow Control Insert

Pressure Compensated Fixed Flow Insert

Description	order code	pack code
VSCF01— 2 l/min	VSC140-1	1
VSCF02— 2.5 l/min	VSC140-2	2
VSCF03— 4 l/min	VSC140-3	3
VSCF04— 5 l/min	VSC140-4	4
VSCF06— 6.5 l/min	VSC140-6	6
VSCF08— 7.5 l/min	VSC140-8	8
VSCF11— 9.5 l/min	VSC140-11	11
VSCF16— 12 l/min	VSC140-16	16





C12D

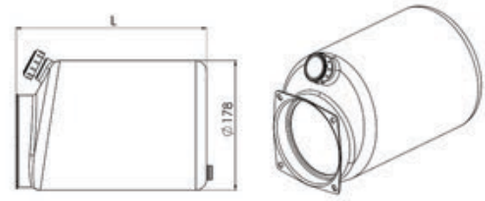
N

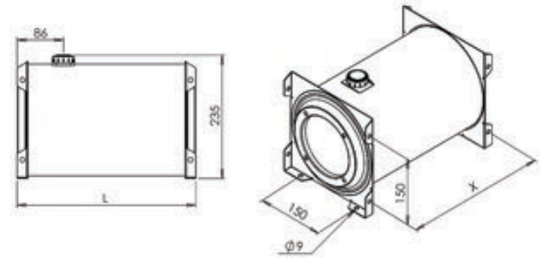
Steel Oil Tank—black powder coated

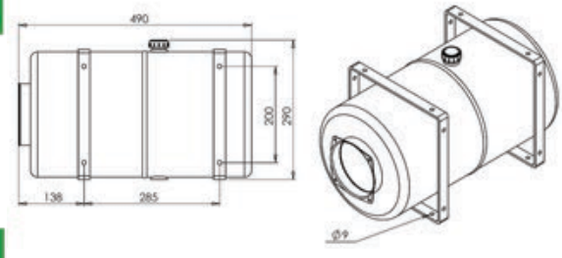
Cylindrical

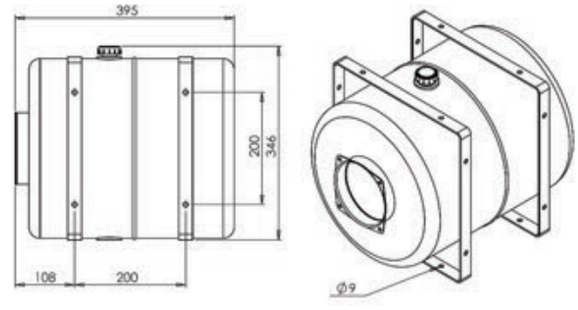
Description	*Length*	*Order code*	*Pack code*
1 Litre Steel	139	HP150002-1	01S
2 Litre Steel	218	HP150002-2	02S
4 Litre Steel	235	HP150002-4	04S
5 Litre Steel	270	HP150002-5	05S
8 Litre Steel	380	HP150002-8	08S
10 Litre Steel	340 (X=322)	HP150026-10	10S
15 Litre Steel	490 (X=472)	HP150026-15	15S
23 Litre Steel	490	HP150026-23	23S
30 Litre Steel	395	HP150026-30	30S
Weld Collar		HP150002-0WN	NWC

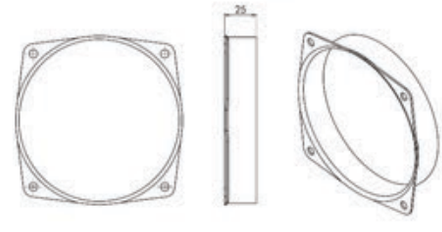












05P

Accessories

Description	*Order code*
3/4" Plastic Filler/Breather	K5630231034
1" Plastic Filler/Breather	K0457252100

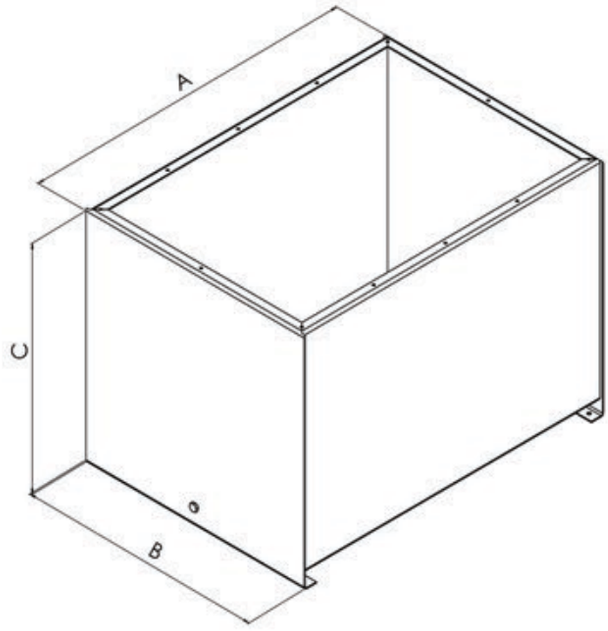

1" BSP


3/4" BSP

Steel Oil Tank

Rectangular

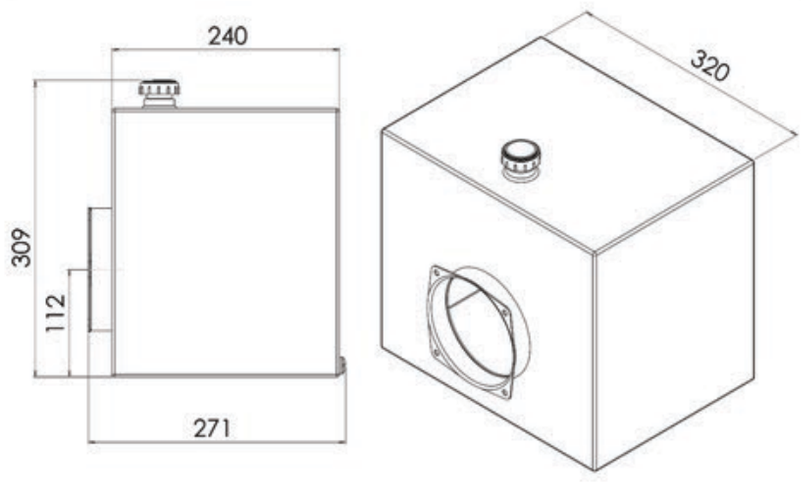
Description	A	B	C	order code	pack code
12 Litre	335	270	290	CF-12	12C
16 Litre	410	325	305	CF-16	16C
30 Litre	470	375	335	HP4306	30C
55 Litre	600	470	365	HP050101	55C



Powerpacks are fitted to the lid of these tanks. For details on the position and orientation please contact our sales team.

These tanks are finished in zinc rich primer as standard.

Description	order code	pack code
18 Litre Steel	HP4370	18S

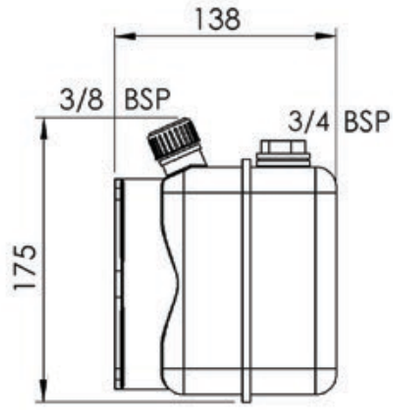


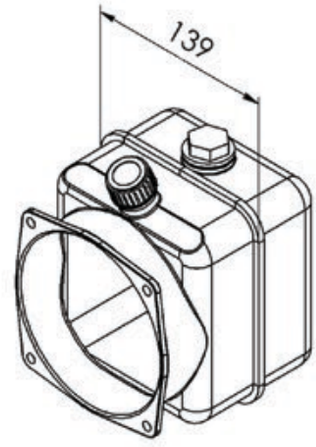
05P

Plastic Oil Tank

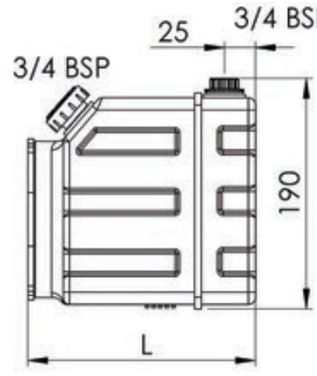
Rectangular

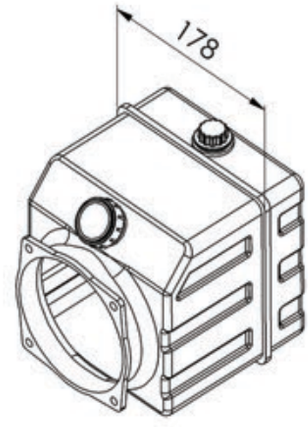
Description	order code	pack code
1.5 Litre Plastic	HP150001-1D	01P





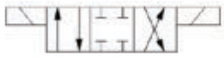
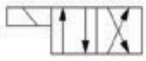
Description	Length(L)	order code	pack code
3 Litre Plastic	190	HP150001-3	03P
5 Litre Plastic	285	HP150001-5	05P
8 Litre Plastic	432	HP150001-8	08P

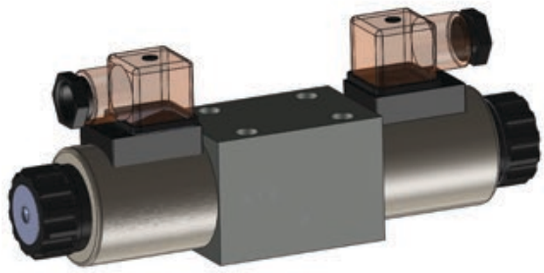




05P

Stacking Plates

Valve diagram	Part Number	Code
	4WE6J60	CD
	4WE6H60	CD
	4WE6G60	CD
	4WE6E60	CD
	4WE6C60	CS
	4WE6Y60	CS
	4WEHB60	CS
Lever operated valves		CL



Voltages available 12VDC, 24VDC, 24VAC, 110VAC, 220VAC
Push and Twist Manual Overrides also available

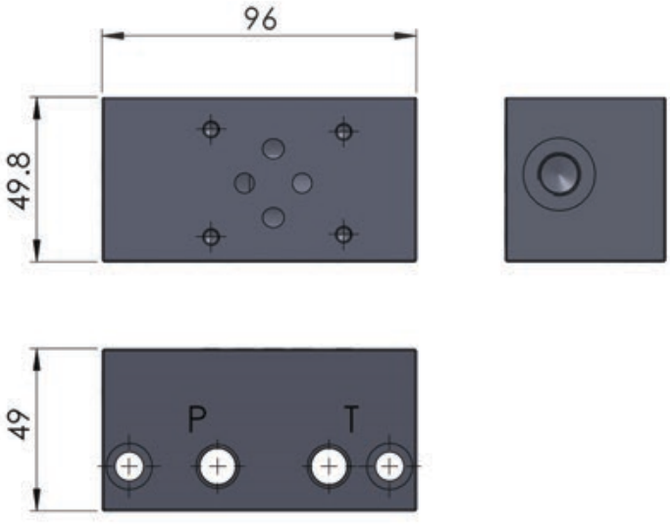
Working Pressure (Max)	Flow (Max)
315 bar	40l/min

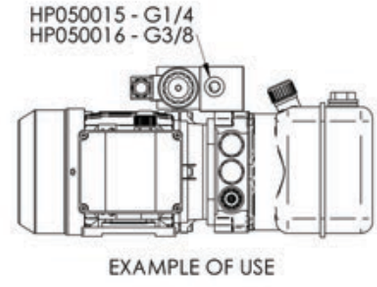
NG6 Valves— Valve configurations and control voltages to be stated separately.

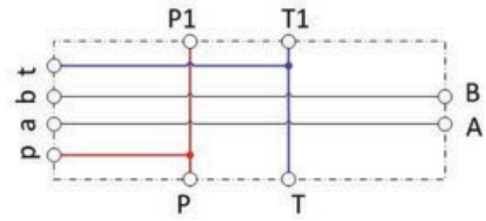
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Stacking Plates

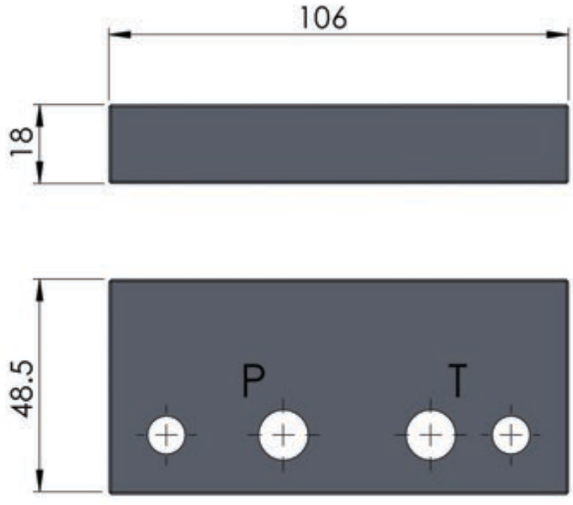
Description	Ordering code
CETOP 3 Parallel Adaptor Block - 1/4	HP050015
CETOP 3 Parallel Adaptor Block - 3/8	HP050016

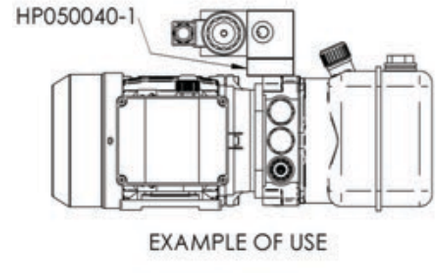


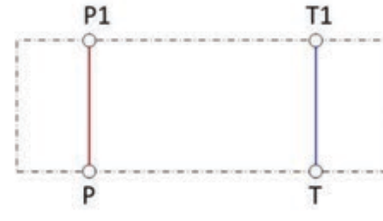




Spacer Block - 18mm	HP050040-1



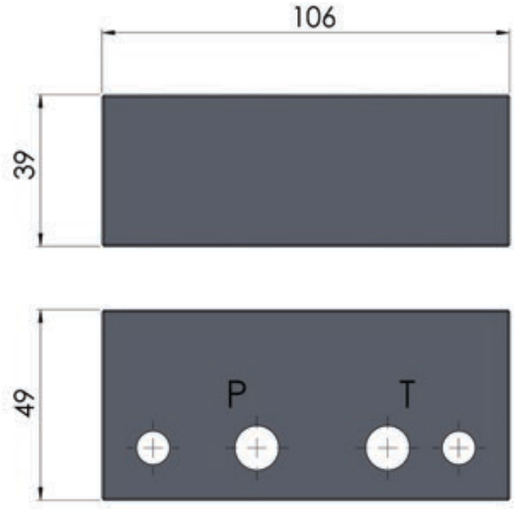


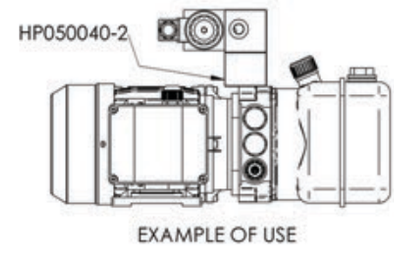


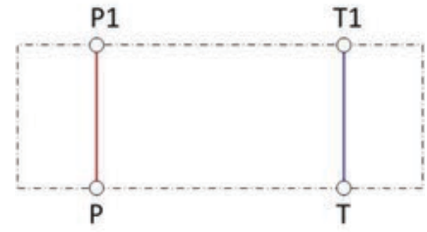
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Stacking Plates

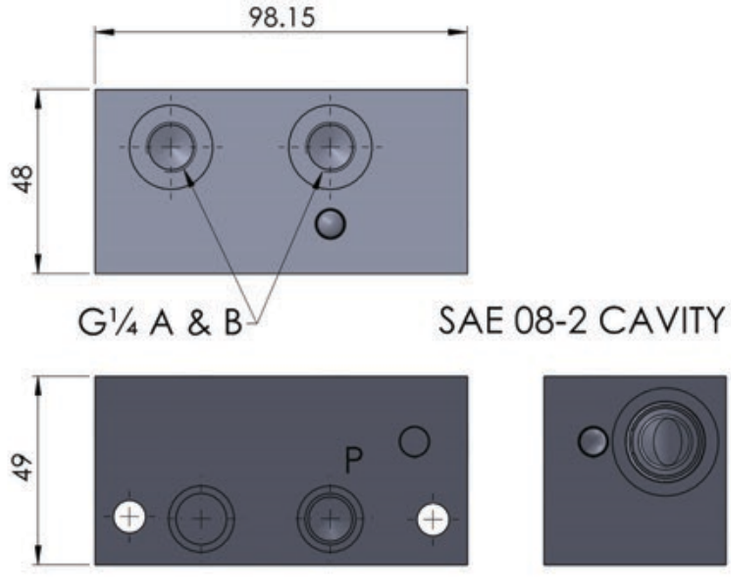
Description	Ordering code
Spacer Block - 39mm	HP050040-2

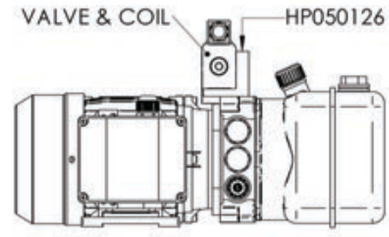


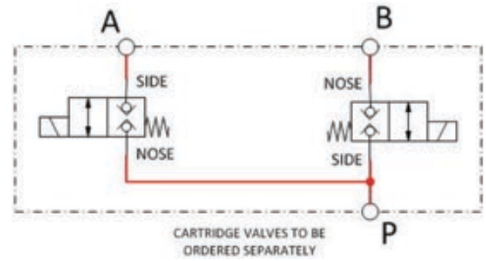




Dual Single Acting Adaptor Block	HP050126



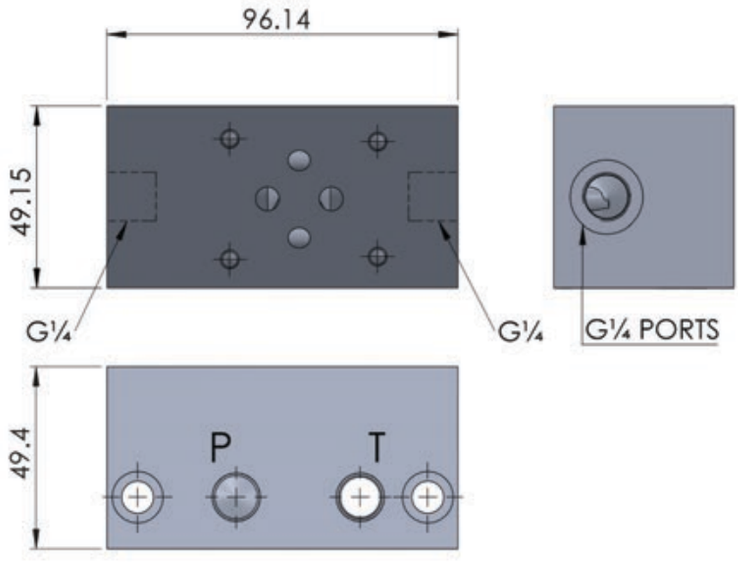


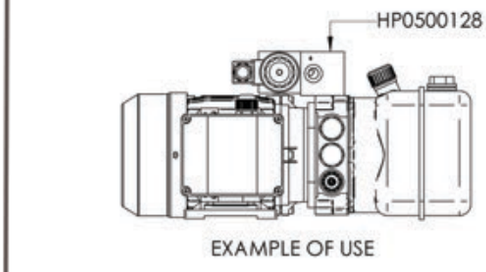


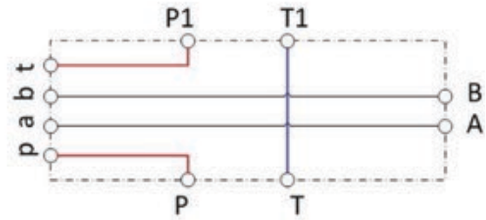
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Stacking Plates

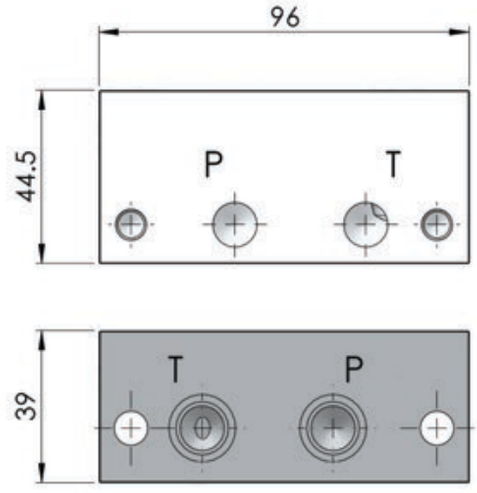
Description	Ordering code
CETOP 3 Series Adaptor Block	HP050128

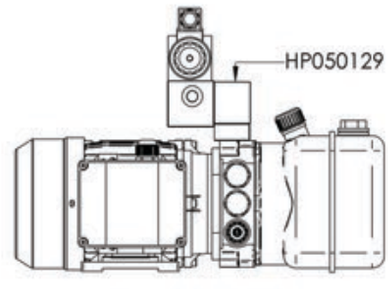


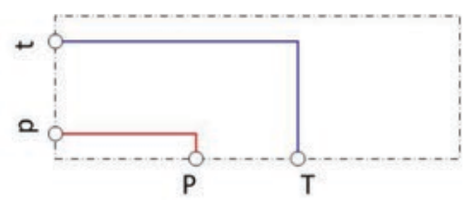




Horizontal to Vertical Adaptor	HP050129



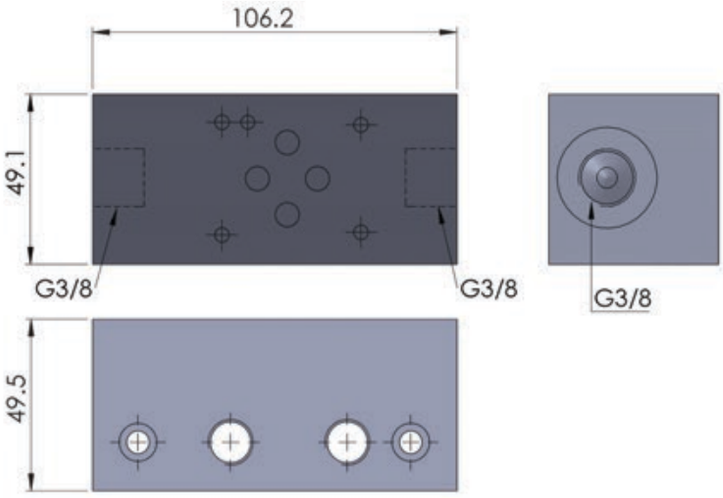


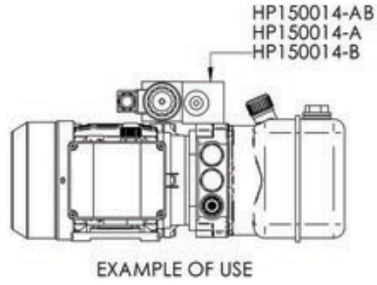


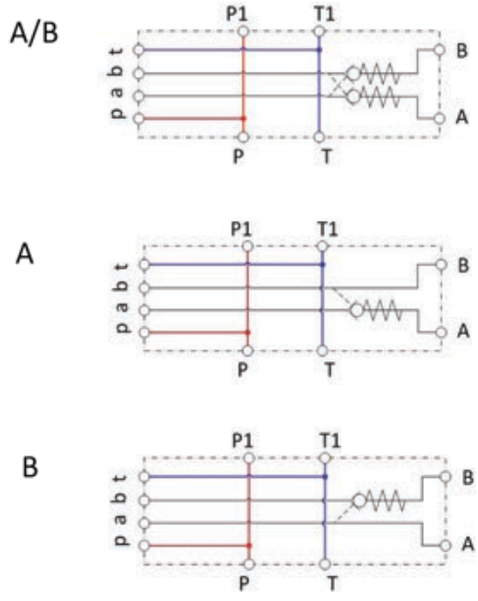
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Stacking Plates

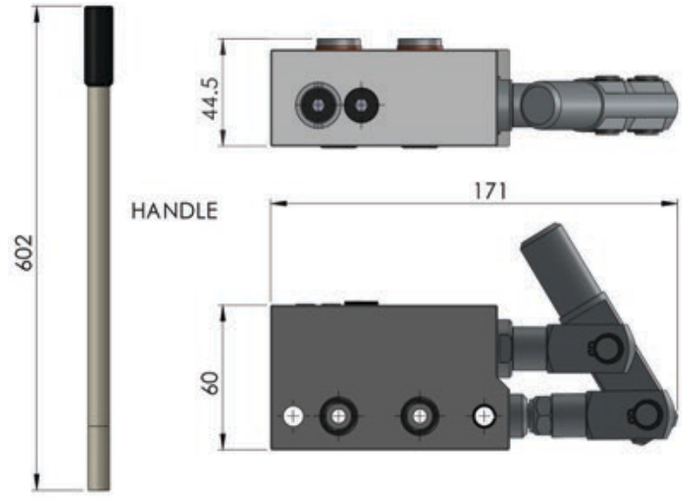
Description	Ordering code
P.O. Check Adaptor on A and B	HP150014-AB
P.O. Check Adaptor on B Only	HP150014-B
P.O. Check Adaptor on A Only	HP150014-A

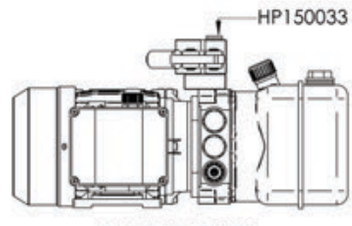


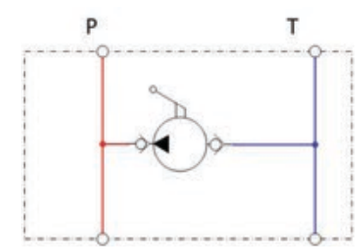




Manual Pump 8cc	HP150033



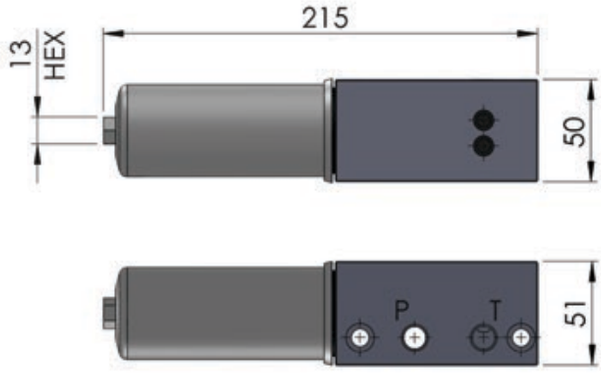




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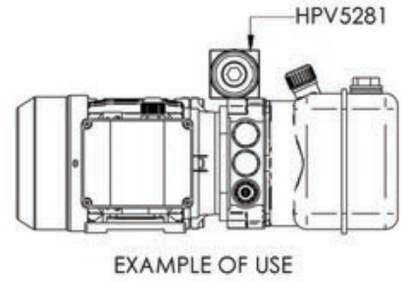
Stacking Plates

Description	Ordering code
Modular Return Filter	HPV5281



15mm nominal

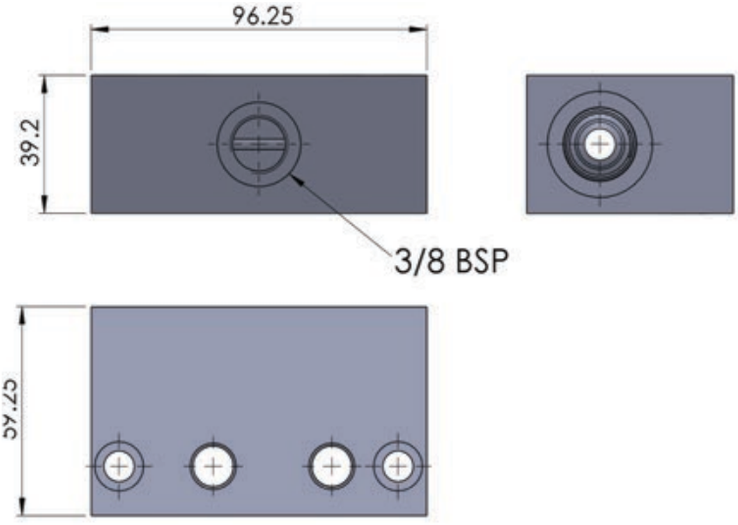
15 L/Min Max

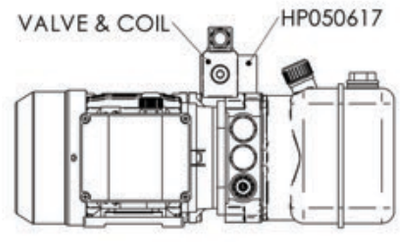


EXAMPLE OF USE

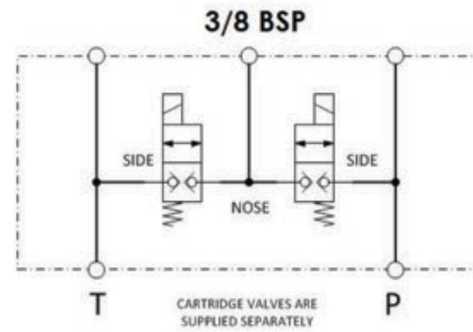


Single Acting Valve Block	HP050617



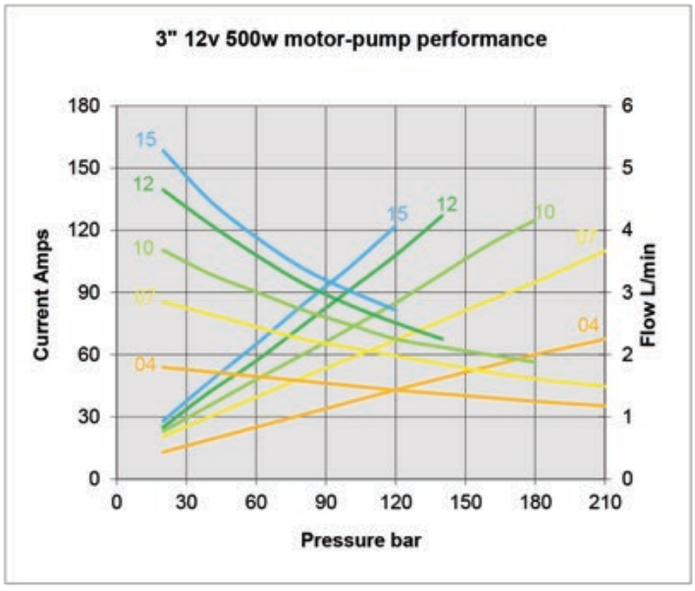


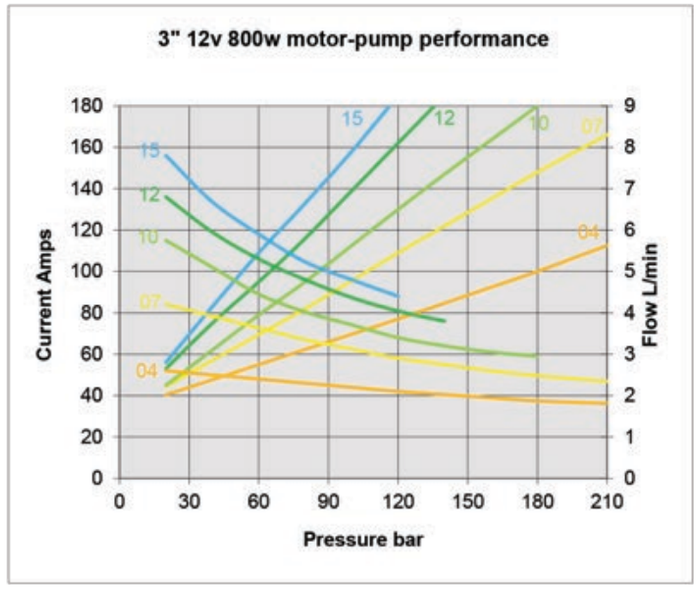
EXAMPLE OF USE

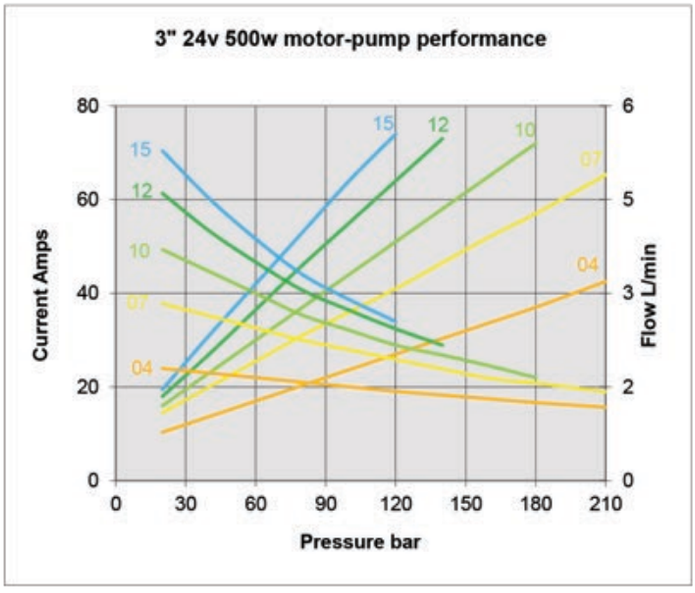


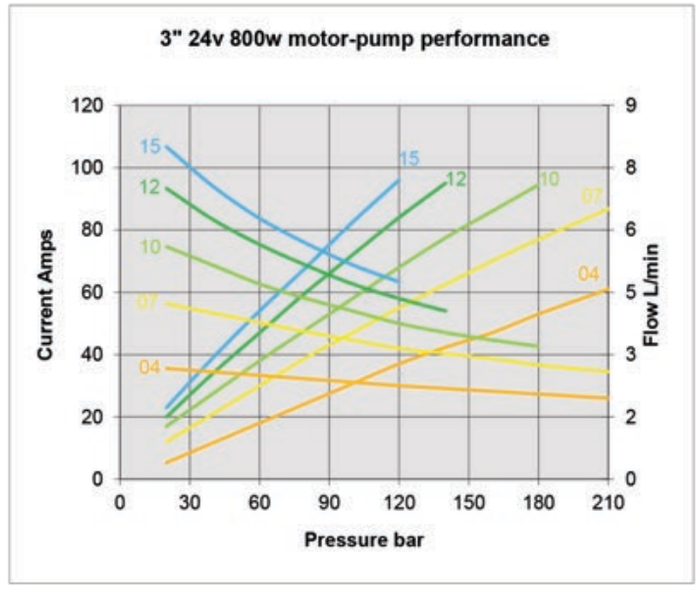
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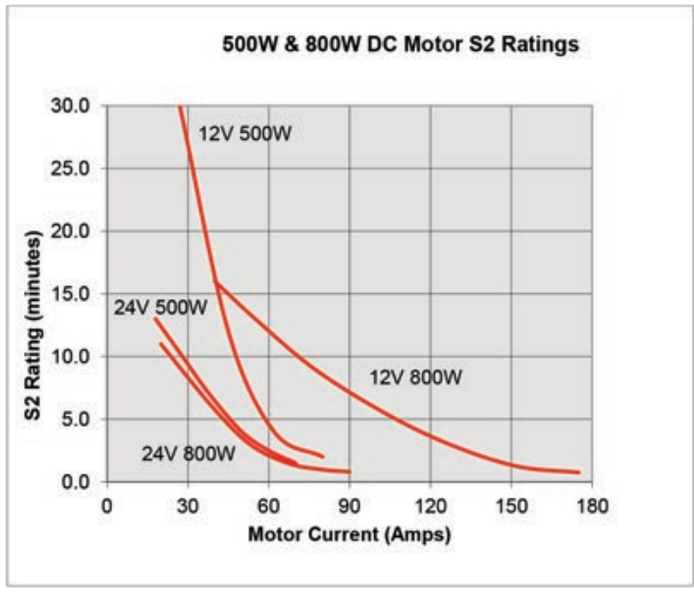
Nominal motor-pump performance at constant voltage and at nominal 20-25° C with ISO 3448/VG22



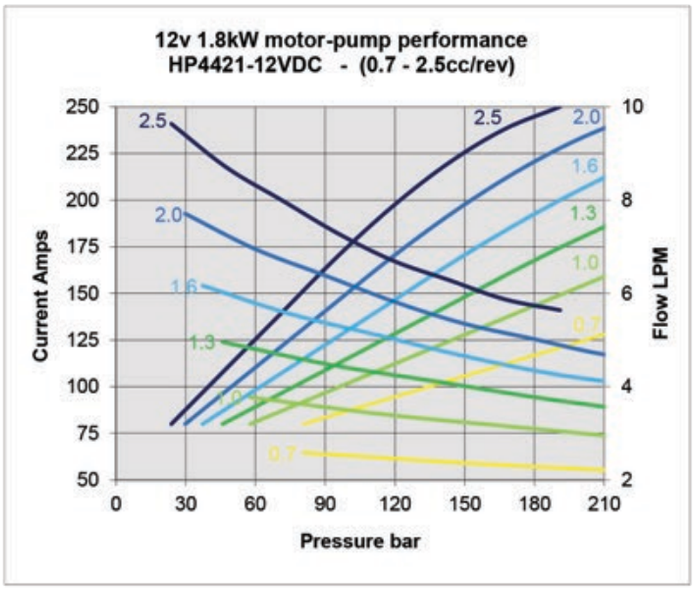


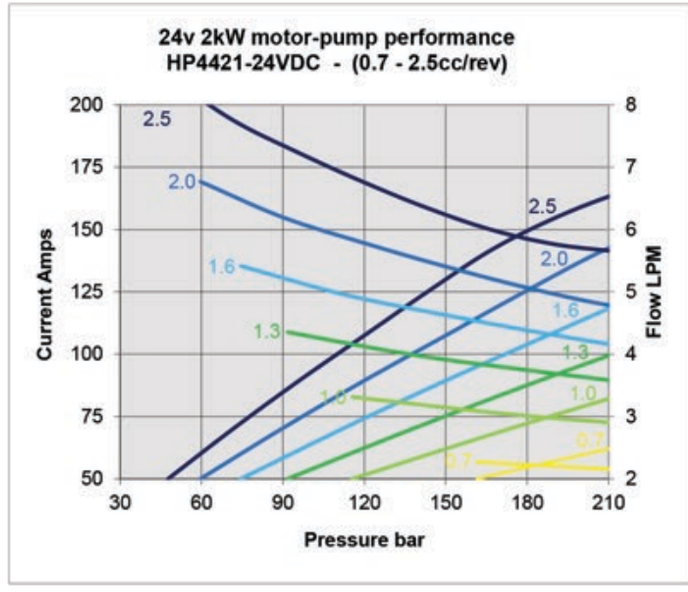


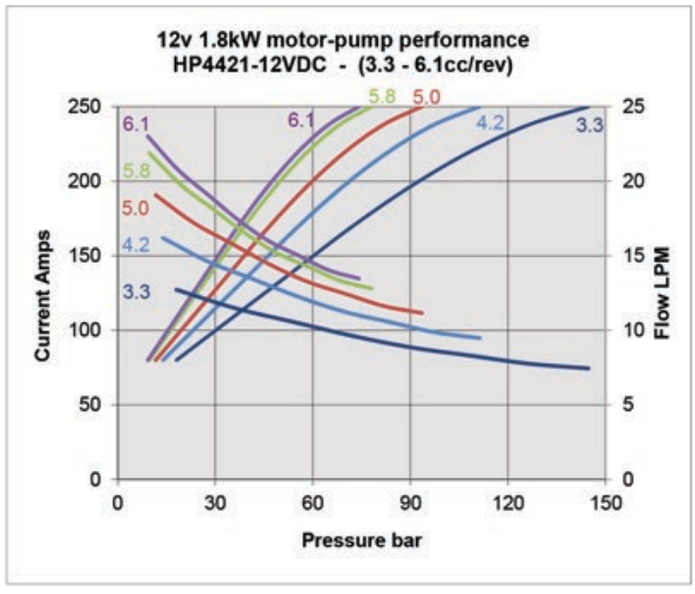


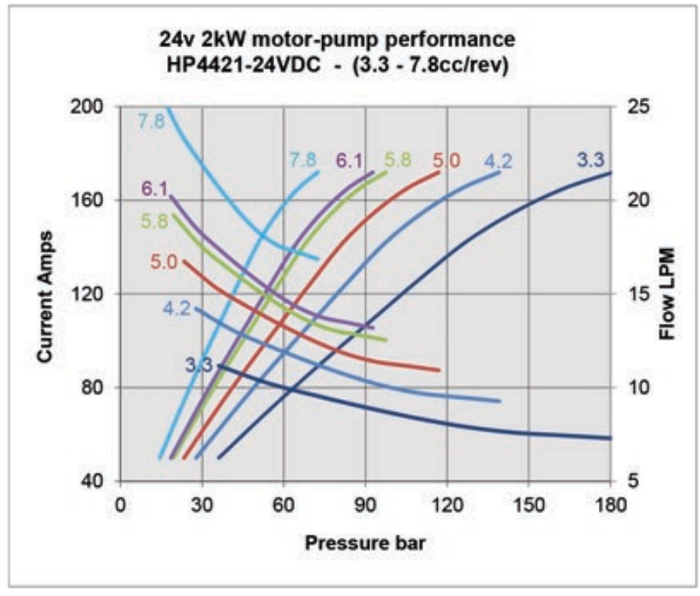


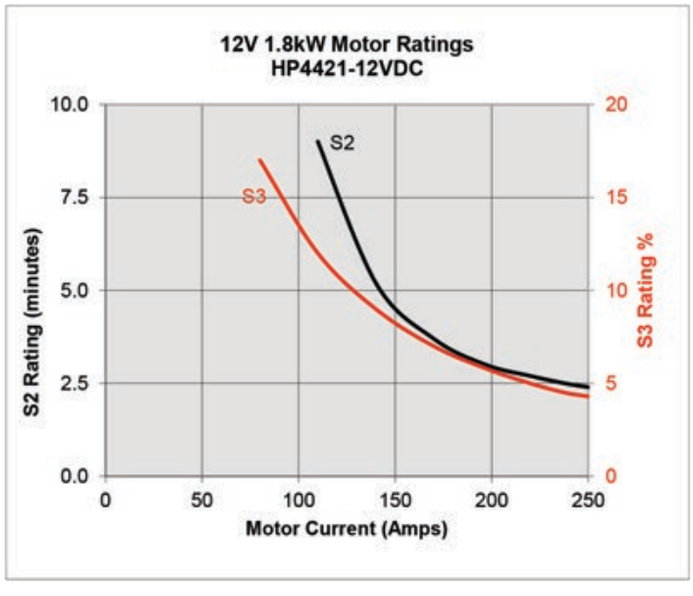
Nominal motor-pump performance at constant voltage and at nominal 20-25° C with ISO 3448/VG22

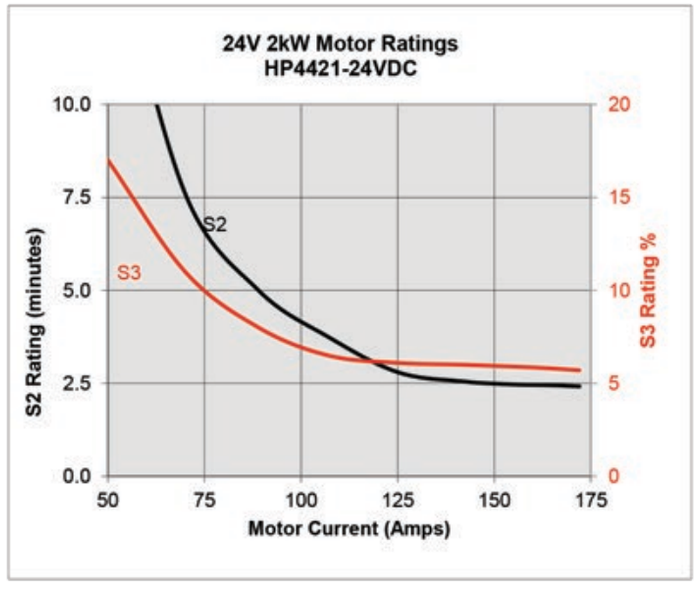




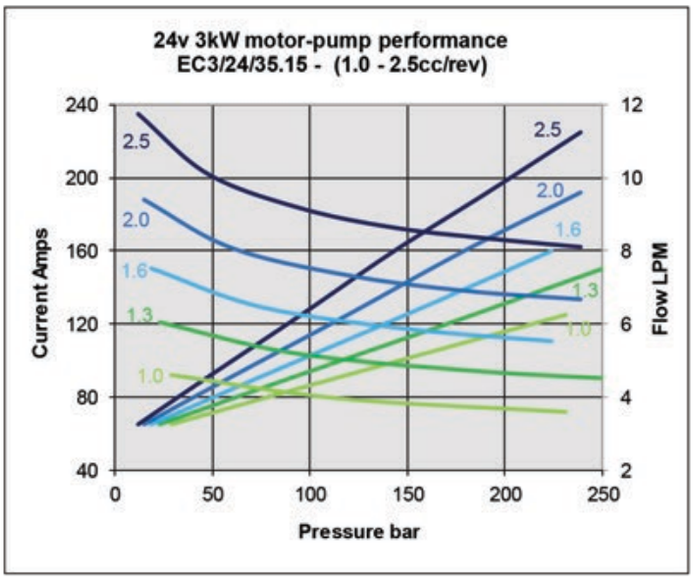


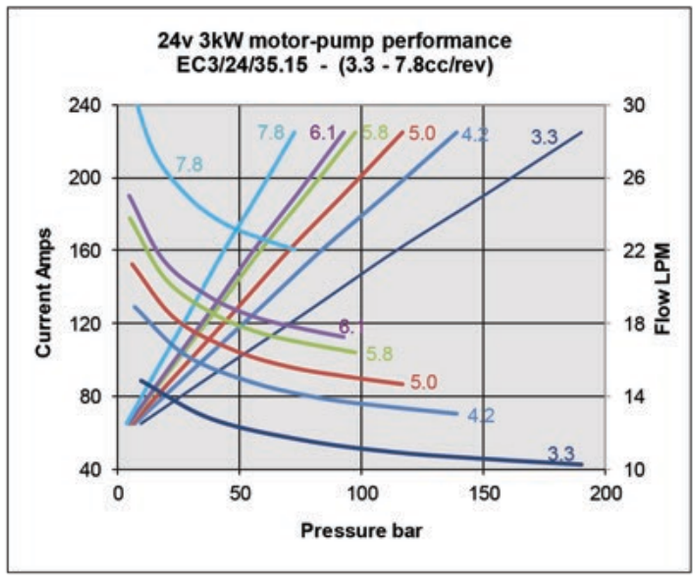


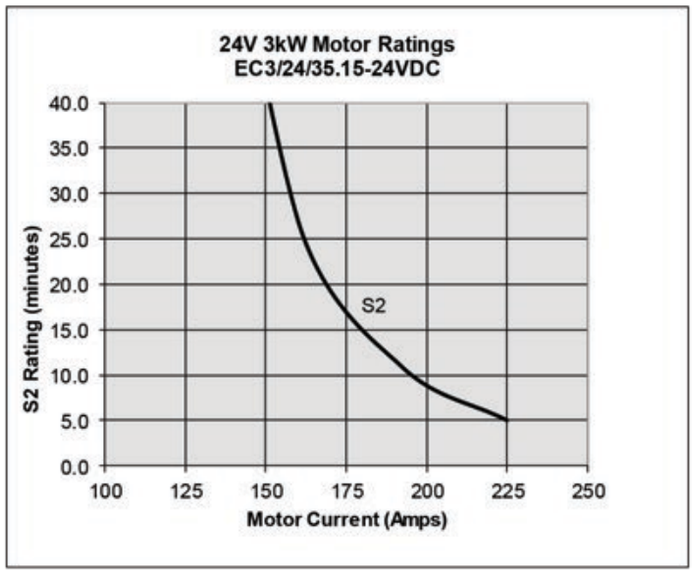




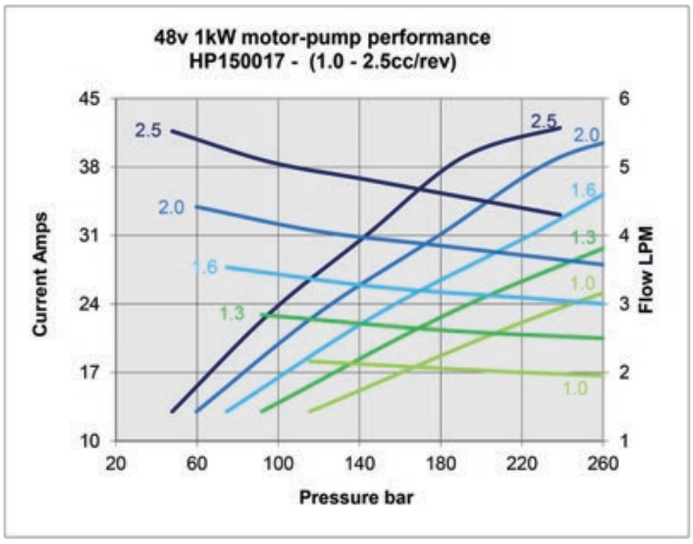
Nominal motor-pump performance at constant voltage and at nominal 20-25° C with ISO 3448/VG22

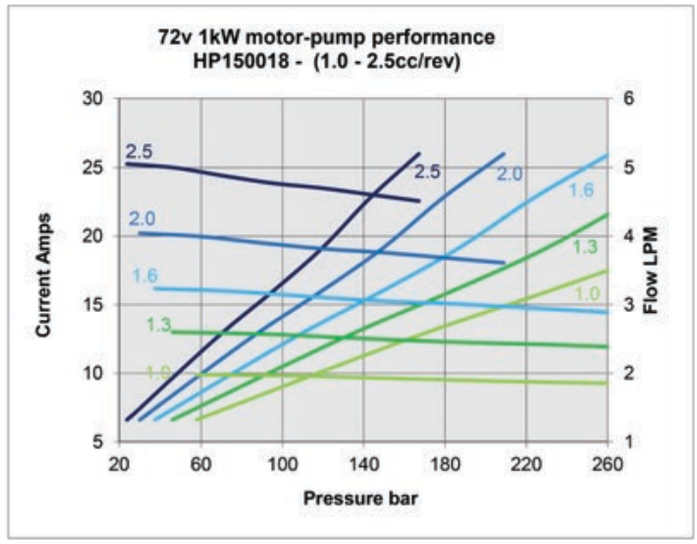


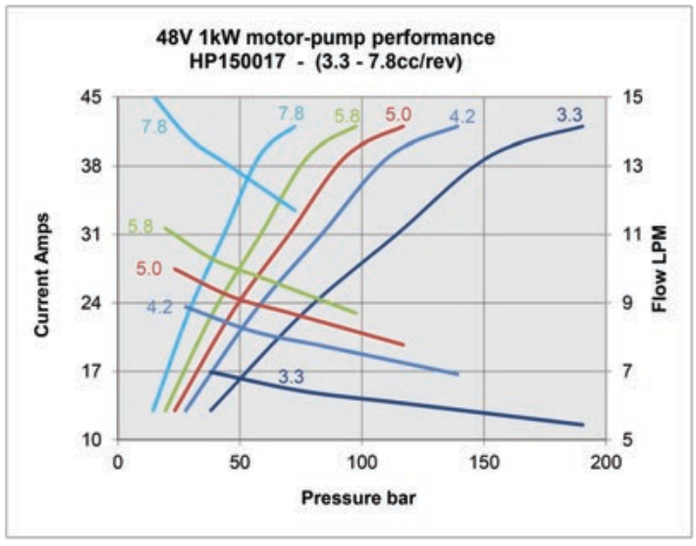


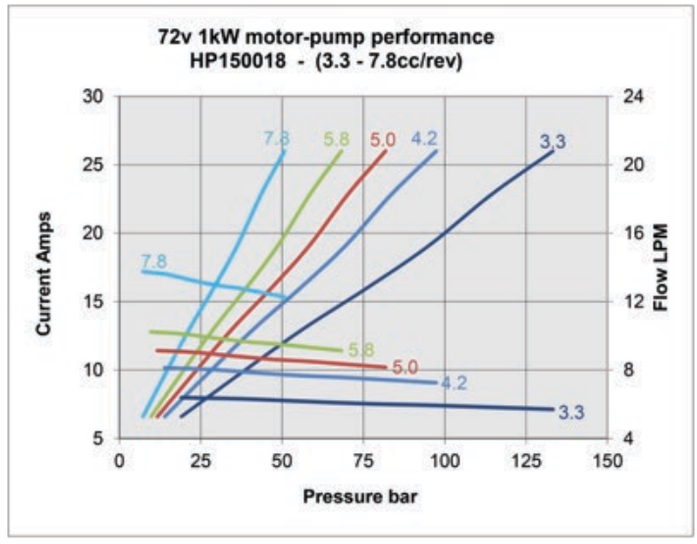


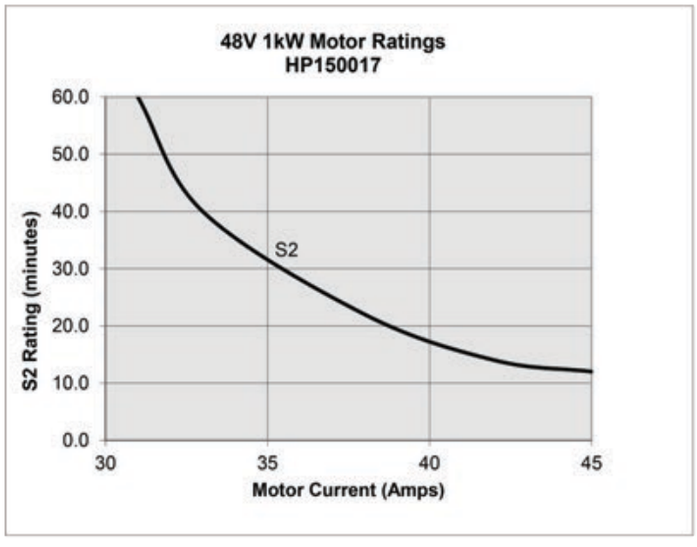
Nominal motor-pump performance at constant voltage and at nominal 20-25° C with ISO 3448/VG22

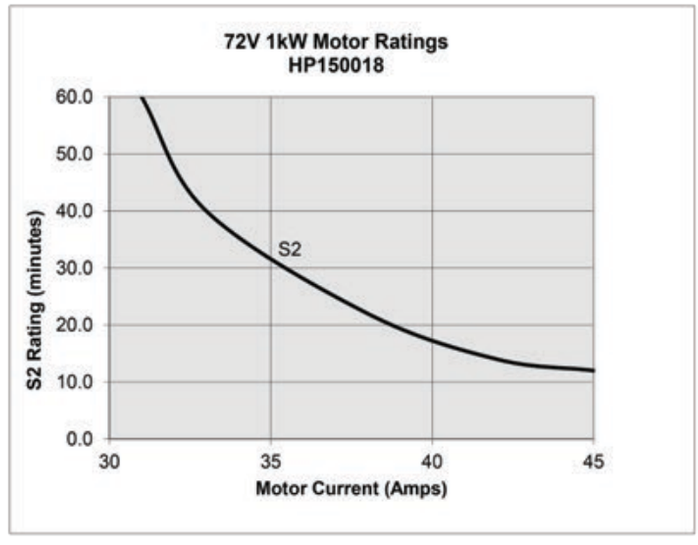




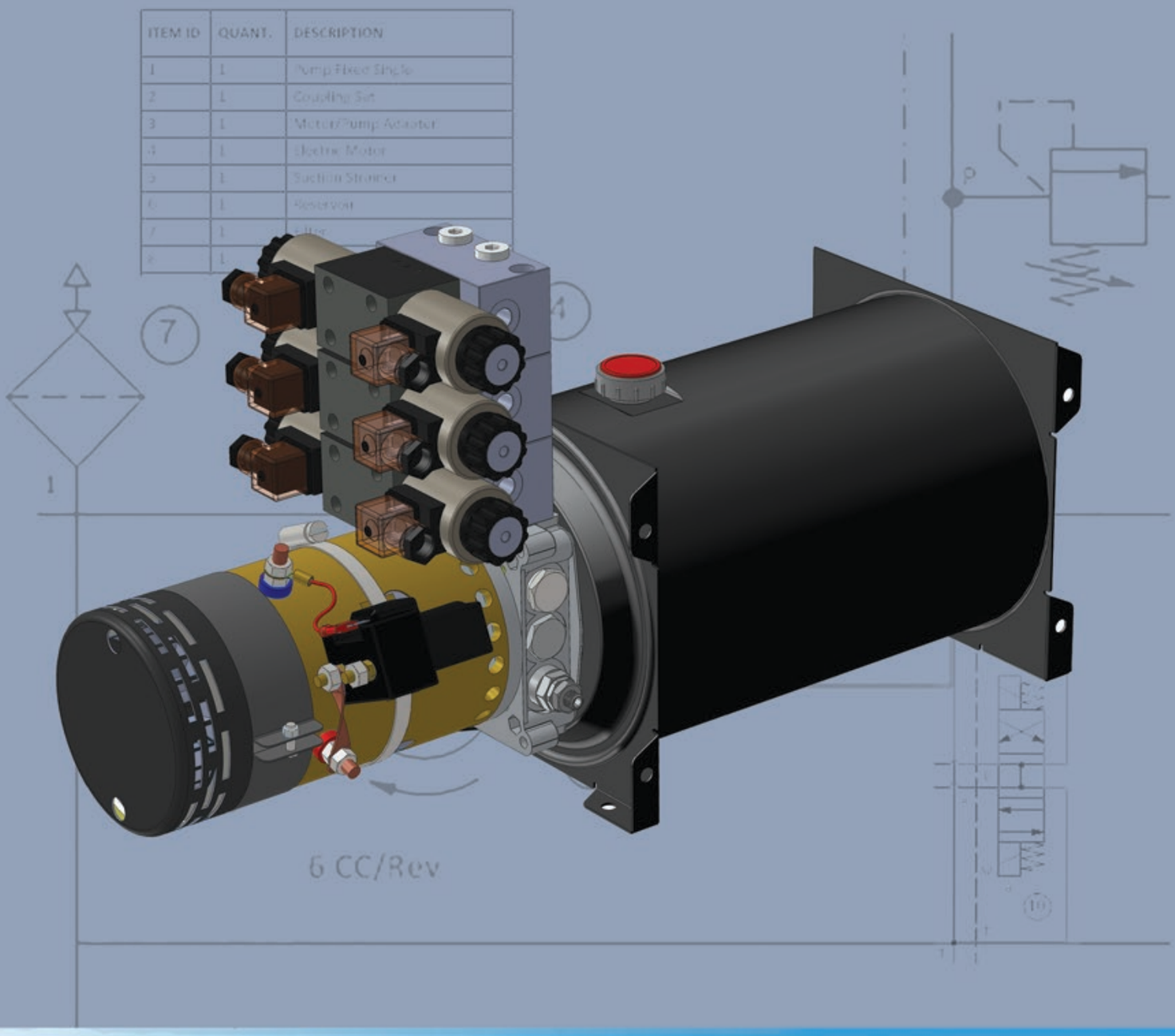








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Our company has established itself as an innovative manufacturer of the Micro Hydraulic power pack with OEM customers throughout the UK and with Distributors Worldwide.

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